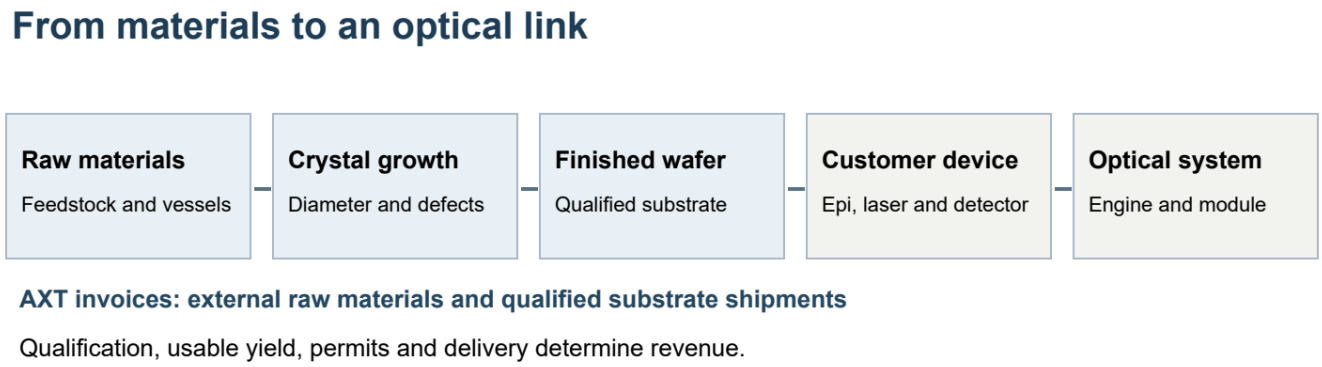
AXT Inc. (AXTI)

Full investment review. November 4, 2026. Framework 8.0. Nasdaq common shares. Bottleneck; covered company; holding status unavailable in the saved record.

The Q4 2027 calculated ceiling is USD 73.75 per share. It values Q4 2027 through Q3 2028 at 20 times parent-economic operating profit and the existing 68 million share proxy.

Analytical score: 45/60, equivalent to 75/100. Protected Excel Base: 75. Individual outcome probabilities are separate judgments. Monetary tables use USD millions; share counts use millions; valuation tables use USD per share.

AXT's opportunity combines stronger InP demand, higher accepted output and a richer product mix. The forecast also includes Tongmei equity dilution, lower-tier minority claims, later pricing normalization, capital spending, customer shipment credits and contingent redemption cash payments.



Explanatory business-model diagram. It identifies the operating boundary and contains no additional market forecast.

1. Thesis and the economic boundary

AXT grows compound-semiconductor crystals and supplies finished substrates for optical, wireless and sensing devices. The operating chain runs from raw materials through crystal growth, slicing, polishing and defect control, then into customers' epitaxy, lasers, detectors, optical engines and modules.

Qualified low-defect material is the scarce input being assessed. AXT earns substrate and consolidated external raw-material revenue. The model assigns each customer and production opportunity to that invoice boundary.

Furnace productivity, wafer diameter, usable yield, permit coverage and customer acceptance determine output. Larger theoretical area creates value only when the material meets the relevant defect and qualification specifications.

The current classification is a bottleneck exposed to capacity response. Rival qualification, downstream integration and export policy can change its duration.

2. Supercycle, S-curve and market expansion

AI data-center optical links provide the immediate demand cycle. Higher-speed pluggables, silicon-photonics light sources and later CPO/NPO adoption can increase material area and quality requirements. These architectures share customer and substrate pools.

The established InP business is in a volume ramp. Six-inch InP is earlier in qualification and pilot manufacturing. GaAs, germanium and external raw materials provide different end-market exposure.

The expansion lens assigns full thematic relevance to near-term optical demand, a 50% thematic weight to adjacent physical-AI/sensing opportunities, and 20% to more distant uses. These assessment weights are separate from the numerical delivery and funding probabilities.

3. Analytical score and risk allocation

The seven categories sum to 45/60. The protected workbook Base remains 75. This score measures thesis quality; the financial calculation uses the explicit operating and funding branch probabilities.

Five deductions cover export/geopolitical risk, capacity/yield/capex execution, customer concentration and overlapping reservations, competition/substitution, and ownership/redemption/dilution.

Category	Points	Maximum	Current evidence
Core thesis	17	20	Qualified crystal growth, defect control and customer reservations
Cycle	9	10	Rapid InP demand and output ramp
TAM expansion	9	10	High-speed optics plus adjacent materials applications
Catalysts	5	5	Q4 execution, December financing, 2027 ramp
Macro	7	10	AI demand balanced by export and policy exposure
Alpha	3	5	Useful technical and local-language detail; unknown unit economics
Risks	-5	0	Five risk classes deducted
Total	45	60	75/100

4. Latest reported quarter

For the three months ended September 30, 2026, revenue was 73.410, gross profit 41.095 and GAAP operating profit 28.427. R&D was 3.944 and SG&A 8.724. The resulting GAAP gross margin was 55.9801%.

Below operating profit, net interest was 5.895, equity income 0.237 and other income -0.238. Pretax income was 34.321 and tax 4.571. Consolidated net income was 29.750, with 4.027 allocated to noncontrolling interests and parent net income of 25.723.

Reported basic and diluted weighted-average shares were 63.827 and 65.862. Stock compensation was 1.069, including 0.065 in cost of goods sold and 1.004 in operating costs.

Rounded product disclosure totals 73.444: InP 54.3, GaAs 9.1, germanium 0.344 and raw materials 9.7. For exact total calibration, the reconstruction uses InP 54.266. This residual allocation is explicitly modeled.

Q3 2026 actual	USD million
Revenue	73.410
Gross profit	41.095
R&D	3.944
SG&A	8.724
GAAP OP	28.427
Pretax income	34.321
Book tax	4.571
Parent net income	25.723

5. Guidance and the latest two management calls

The October 29 Q3 call and July 30 Q2 call were reviewed through their visible Q&A and closing. The displayed sources cover management's operating and financing discussion. Original-audio fidelity remains unverified.

Management describes approximately 90 revenue for Q4 under permit-held or permit-exempt routes, approximately 14 GAAP operating costs and 13 non-GAAP operating costs. The forecast uses the GAAP cost basis.

The CFO says the next gross-margin percentage begins with 6 and excludes a high 6. The 66% group gross margin is a model selection inside that wording, assessed together with the earnings guidance. The underlying answer leaves the GAAP/non-GAAP basis implicit.

The displayed guidance has GAAP 0.56-0.58 and non-GAAP 0.57-0.59 income figures alongside 66.1 million shares. Reading these as per-share ranges follows the context; explicit per-share words are absent from the raw sentence.

At the selected Q4 operating and tax inputs, modeled parent common income is 37.264653 and the guidance-denominator EPS diagnostic is 0.563762. The valuation continues to use the sanctioned economic share schedule.

6. Q4 product calibration

Q4 reference revenue is 90.000: core InP 65.800, six-inch InP 4.200, GaAs 9.400, germanium 0.350 and raw materials 10.250.

To reconcile 66% group gross margin after the selected other-product margins, core InP gross margin is 78.015957%. This is a LOW-confidence residual calibration. It does not establish a disclosed core-product margin.

Six-inch InP starts at 6% of InP revenue and 50% gross margin. The mix is a revenue allocation. Greater wafer diameter and modeled ASP changes remain separate from this mix share.

The following table contains the exact product GP arithmetic behind the group anchor.

Q4 product	Revenue	Gross margin %	Gross profit
core InP	65.800	78.016	51.335
six inch InP	4.200	50.000	2.100
GaAs	9.400	32.000	3.008
Ge	0.350	25.000	0.088
raw materials	10.250	28.000	2.870
Total	90.000	66.000	59.400

7. Technical context: area, yield and defect requirements

The retained expert page discusses emitter area and quality requirements. Its 18-light-source example applies to a particular architecture. Device yield, substrate finishing yield and module acceptance have separate denominators.

The source is an anonymous expert interview published by P Equity Research on October 19. The author interpretation and expert comments share one source chain.

Q2. How is the InP substrate consumption of a single optical module calculated?

Optical modules are mainly divided into two technical routes, EML and CW. Taking a 3-inch substrate as an example, without considering downstream epitaxy and optical chip yield, one substrate can be cut into about 10,000 EML chips for 100G and 200G. But once downstream yield is included, the effective chip output per wafer is a little over 2,000. For conventional 70 mW and 100 mW CW chips, one 3-inch substrate can be cut into about 8,000 dies, and after yield the effective output is likewise a little over 2,000. Although CW chips have fewer dies per wafer, their overall yield is usually slightly higher than EML's.

Q3. How will CPO solutions affect demand for InP substrates?

CPO is currently at the sample stage and not yet in large-scale volume production. CPO will drive significant growth in InP substrate demand, mainly through die area and quality requirements. **A CPO module is expected to use 18 CW light sources**, with power requirements far above conventional products. For example, the 200 mW and 300 mW CW chips being developed by Yuanjie Semiconductor have a single bare die area 2 to 3 times that of today's 70 mW and 100 mW chips, which directly increases substrate area demand. CPO modules also place stricter demands on substrate quality. Because continuous emission from CW lasers generates serious heat, high defect density raises the probability of chip failure. Therefore, for 300 mW CW sources used in CPO, **the maximum substrate defect density is required not to exceed 200**, whereas for orders for conventional 100G and 200G optical modules or 70 mW and 100 mW CW sources, **the average defect density requirement is only below 500**.

Q4. How will market supply meet the demand growth expected in 2027?

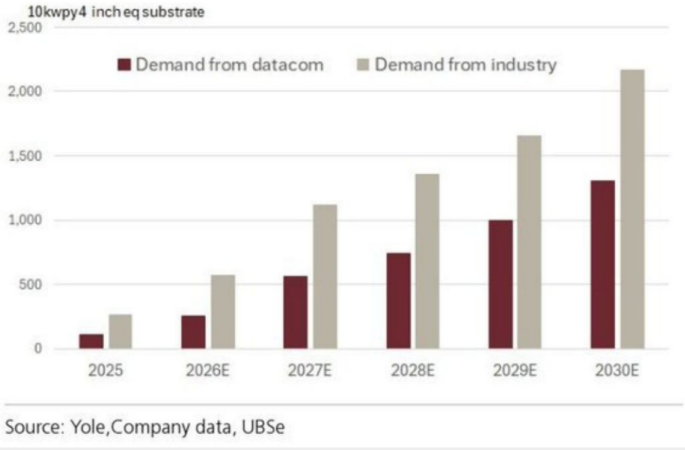
P Equity Research InP Expert Call, October 19, 2026, retained page 7. Anonymous expert interpretation of emitter area, downstream yield and defect requirements. The architecture example has a specific scope.

8. Counter-view on supply and pricing

The expert anticipates easing supply and pricing later in 2027; the publisher emphasizes qualification and yield constraints. Both interpretations inform the outcome distribution.

The source mixes wafer-equivalent definitions and an unresolved chart-axis multiplier. The forecast uses a normalized accepted-output index and keeps these source capacity figures qualitative.

Figure 25: InP substrates demand forecast



Source: Yole,Company data, UBSe

Q6. What is your view of the capacity expansion plans of major manufacturers such as AXT and Yunnan Germanium, and their effect on market supply next year (2027)?

The capacity targets of 200,000 or 300,000 wafers mentioned by AXT and Yunnan Germanium should be long-term plans. By 2027, the limit of their capacity expansion is roughly double 2026 levels. Even so, AXT's monthly capacity could then reach nearly 100,000 wafers, with annual capacity above 1 million. Yunnan Germanium and Xiandao will not be small either; together the three are expected to achieve annual capacity of more than 3 million wafers in 2027. Therefore, by the second half of 2027, the tight market supply is expected to ease.

Q7. How have InP substrate prices changed recently, and how should future price trends be judged?

Prices have risen slightly recently, by about 5%, but far less than the rise in the first half of 2026. Specifically, 2-inch products were around RMB 1,500 last month (September 2026); 3-inch products are close to RMB

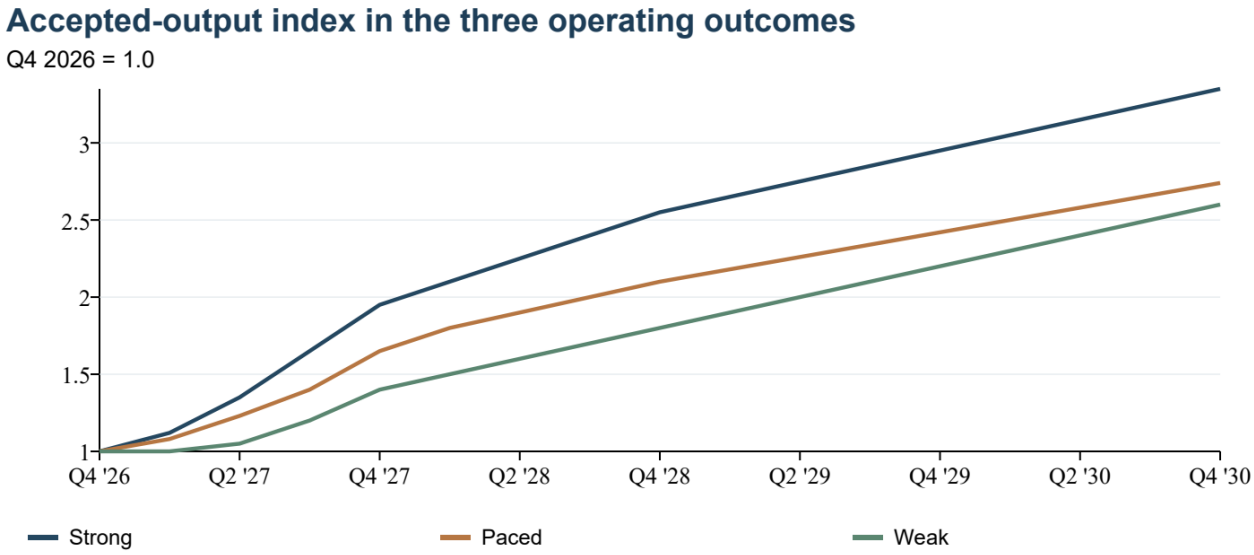
P Equity Research InP Expert Call, October 19, 2026, retained page 9. Expert supply and pricing view. The chart axis and mixed wafer-equivalent units remain unresolved.

9. Accepted output and physical execution

Management describes end 2026 physical capacity at 2.75-3.0 times end 2025, followed by approximately another doubling by end 2027. These are capacity reference points with different starting dates.

The model normalizes Q4 2026 accepted InP output to 1.0. Strong, paced and weak end 2027 indices are 1.95,1.65 and 1.40. The conversion includes qualification, yield and acceptance through the selected outcome; it stays within the modeled capacity direction.

Earlier 60/130 million exit revenue-capacity figures were described as overtaken in the Q3 call. They supply dated context; the new calculation uses the explicit output and realization paths.



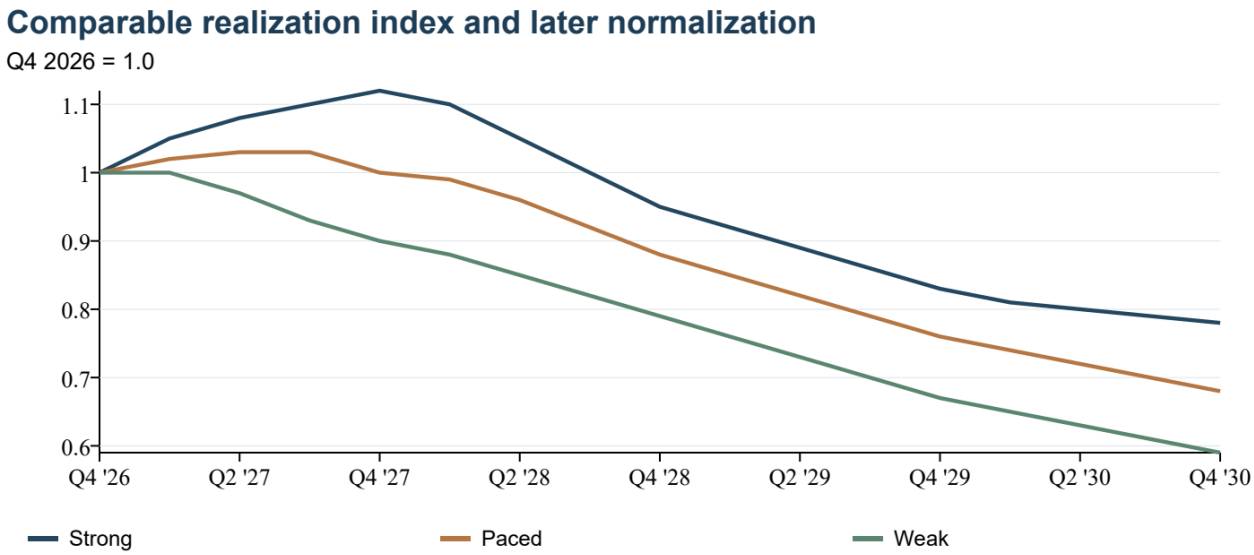
Accepted-output index, Q4 2026 = 1.0. Conditional model estimates; no absolute wafer count is asserted.

Quarter	Strong	Paced	Weak
Q4 2026	1.000	1.000	1.000
Q1 2027	1.120	1.080	1.000
Q2 2027	1.350	1.230	1.050
Q3 2027	1.650	1.400	1.200
Q4 2027	1.950	1.650	1.400
Q1 2028	2.100	1.800	1.500
Q2 2028	2.250	1.900	1.600
Q3 2028	2.400	2.000	1.700
Q4 2028	2.550	2.100	1.800
Q1 2029	2.650	2.180	1.900
Q2 2029	2.750	2.260	2.000
Q3 2029	2.850	2.340	2.100
Q4 2029	2.950	2.420	2.200
Q1 2030	3.050	2.500	2.300
Q2 2030	3.150	2.580	2.400
Q3 2030	3.250	2.660	2.500
Q4 2030	3.350	2.740	2.600

10. ASP, mix and normalization

The strong realization index peaks at 1.12 in Q4 2027, then declines to 0.78 by Q4 2030. The paced and weak paths normalize earlier and more deeply.

Reference InP revenue equals 70 times the accepted-output index times the realization index. A higher mix share does not create another revenue multiplier. Competitive output and customer buffers are included in the paced and weak outcomes.



Comparable realization index, Q4 2026 = 1.0. These are conditional modeled prices and mix effects.

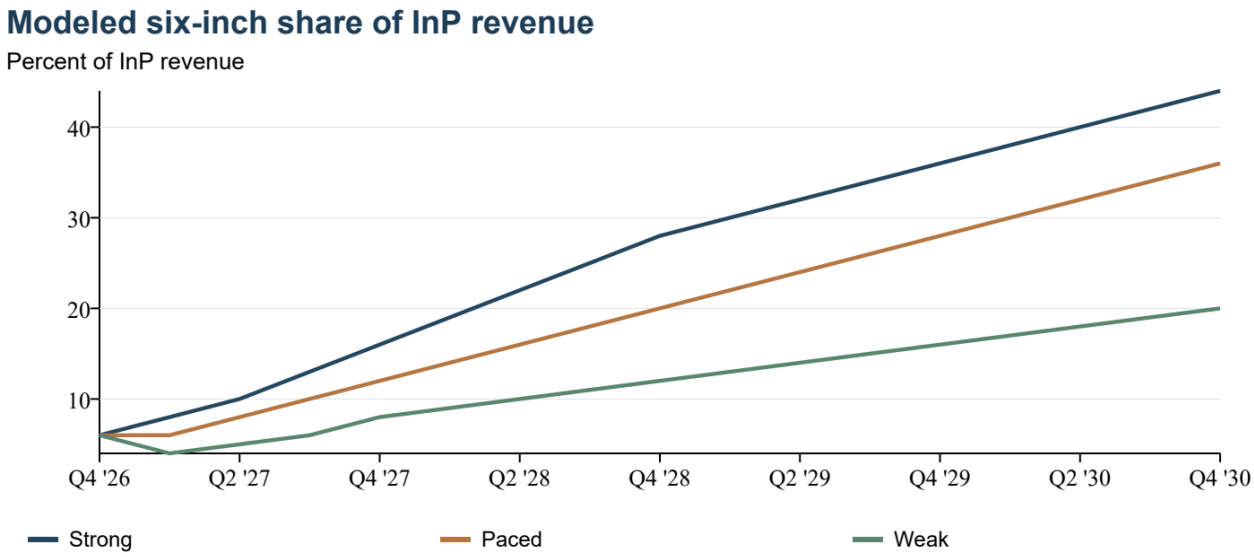
Quarter	Strong	Paced	Weak
Q4 2026	1.000	1.000	1.000
Q1 2027	1.050	1.020	1.000
Q2 2027	1.080	1.030	0.970
Q3 2027	1.100	1.030	0.930
Q4 2027	1.120	1.000	0.900
Q1 2028	1.100	0.990	0.880
Q2 2028	1.050	0.960	0.850
Q3 2028	1.000	0.920	0.820
Q4 2028	0.950	0.880	0.790
Q1 2029	0.920	0.850	0.760
Q2 2029	0.890	0.820	0.730
Q3 2029	0.860	0.790	0.700
Q4 2029	0.830	0.760	0.670
Q1 2030	0.810	0.740	0.650
Q2 2030	0.800	0.720	0.630
Q3 2030	0.790	0.700	0.610
Q4 2030	0.780	0.680	0.590

11. Six-inch qualification and revenue share

The six-inch share follows a separate revenue-mix trajectory. It reaches 16%,12% and 8% in Q4 2027 across strong, paced and weak outcomes.

The current source identifies pilot status and lower initial yield. Exact diameter-resolved shipments, customer acceptance and realized margins remain unavailable. Revenue allocation and modeled gross margins carry LOW confidence.

The framework's dated global-module estimates are USD 67.7 B/131.4 B/148.5 B for 2026/27/28. AXT's addressable substrate pool requires a separate material-content and qualified-supply conversion.



Six-inch share of modeled InP revenue, percent. The shares are assumptions.

Quarter	Strong	Paced	Weak
Q4 2026	6.000	6.000	6.000
Q1 2027	8.000	6.000	4.000
Q2 2027	10.000	8.000	5.000
Q3 2027	13.000	10.000	6.000
Q4 2027	16.000	12.000	8.000
Q1 2028	19.000	14.000	9.000
Q2 2028	22.000	16.000	10.000
Q3 2028	25.000	18.000	11.000
Q4 2028	28.000	20.000	12.000
Q1 2029	30.000	22.000	13.000
Q2 2029	32.000	24.000	14.000
Q3 2029	34.000	26.000	15.000
Q4 2029	36.000	28.000	16.000
Q1 2030	38.000	30.000	17.000
Q2 2030	40.000	32.000	18.000
Q3 2030	42.000	34.000	19.000
Q4 2030	44.000	36.000	20.000

12. A competing qualified supply relationship

Yunnan's November 4 announcement reports a five-year AOI framework with expected 180,000 four-inch N-type InP wafers annually and a separate 2027 annual contract for 180,000 wafers andUSD 194.4 million before tax.

The implied 2027 amount isUSD 1,080 per wafer. Export permission, purchase orders and deliveries govern realized quantities. The source establishes a relevant competing supply relationship.

证券代码：002428　　证券简称：云南锗业　　公告编号：2026-054

云南临沧鑫圆锗业股份有限公司

关于签订日常经营重大合同的公告

本公司及董事会全体成员保证信息披露的内容真实、准确和完整，没有虚假记载、误导性陈述或重大遗漏。

特别提示：

1、合同类型及金额：云南临沧鑫圆锗业股份有限公司（下称“公司”）控股子公司云南鑫耀半导体材料有限公司（下称“云南鑫耀”）与客户 Applied Optoelectronics, Inc.（下称“AOI”）签订了《长期供货协议》，AOI 预计于 2027 年至 2031 年期间向云南鑫耀每年采购 18 万片、合计 90 万片 4 英寸 N 型磷化铟衬底，实际价格和交付数量以后续年度合同、采购订单及出口许可证批准情况为准。同时，云南鑫耀与 AOI 签订了《年度合同》，约定 2027 年度 AOI 采购数量为 18 万片，对应金额为 19,440 万美元（不含税），实际金额视出口许可批准数量及采购订单情况而定。

2、生效条件：《长期供货协议》自双方法定代表人或授权代表签字盖章起生效；《年度合同》自双方签署之日起生效，合同履行以取得并维持两用物项出口许可为前提。

3、合同履行对公司的影响：本次《长期供货协议》和《年度合同》的签订，有望为公司磷化铟产品带来潜在的订单需求，有利于拓宽公司的盈利渠道。若合同顺利履行，预计将对公司履约年度的经营成果产生积极影响，具体影响金额及影响的报告期将视出口许可、采购订单和合同履行的具体情况而定，以公司经审计确认的收入为准。

4、重要风险提示：

（1）本次签订的《长期供货协议》为双方基于长期合作而达成的框架性约定，其中每年 18 万片、五年预计 90 万片为 AOI 的预计采购安排，实际价格和交付数量以双方后续签订的年度合同、采购订单及商务部出口许可证批准数量为准。在后期协议履行实施过程中，存在因客户需求变化、政策及市场环境

1

Yunnan Germanium announcement 2026-054, November 4, 2026, page 1. Original Mandarin notice identifies the 2027 contract and conditional framework volumes.

13. Contract scope and annual purchasing

The original Mandarin terms identify the product and 2027-2031 framework period. Future annual prices and actual orders determine later realizations.

The model uses this evidence to support the competitive and normalization outcomes. It assigns no direct USD 194.4 million loss to AXT because the source names no displaced AXT order.

资产:约 USD 860M

经营范围: Semiconductor Manufacturing

Address：13139 Jess Pirtle Blvd., Sugar Land, TX 77478

地址：德克萨斯州舒格兰市杰斯皮特尔大道 13139 号，邮编 77478

AOI 是一家美国光通信企业，1997 年 2 月 28 日在德克萨斯州组建，2013 年 9 月在纳斯达克挂牌上市，股票代码为 AAOI。公司总部位于美国德克萨斯州，主要业务为设计和制造用于数据中心、有线电视及电信市场的光通信产品。

该客户信誉良好，具备较强的履约能力，履约风险可控。经查询，该客户不属于失信被执行人。公司与该客户不存在关联关系。

三、合同主要内容

（一）《长期供货协议》主要内容

1、交易双方：

采购方：Applied Optoelectronics, Inc.

供应商：云南鑫耀

2、协议标的：4 英寸 N 型磷化铟衬底；

3、预计采购数量：AOI 预计每年采购 18 万片，五年预计采购总量 90 万片；实际交付数量以双方后续签订的年度合同、采购订单及商务部出口许可证批准数量为准；

4、采购周期：2027 年 1 月 1 日至 2031 年 12 月 31 日；

5、生效条件：自双方法定代表人或授权代表签字盖章之日起生效；

6、其他条款：协议对交付计划、付款约定、违约条款、争议解决等方面做出明确约定，实际交付数量及安排以年度合同、采购订单和商务部出口许可证批准数量为准。

（二）《年度合同》主要内容

1、交易双方：

买方：Applied Optoelectronics, Inc.

卖方：云南鑫耀

2、合同标的：4 英寸 N 型磷化铟衬底；

3、合同数量及金额：约定数量 18 万片，对应金额 19,440 万美元（不含税），实际金额视出口许可批准数量及采购订单情况而定；

3

Yunnan Germanium announcement 2026-054, November 4, 2026, page 3. Original Mandarin terms identify four-inch N-type InP, expected annual purchases and separate annual contracts.

14. Permit, cancellation and refund terms

Failure to obtain or maintain the required dual-use export authorization permits cancellation of affected orders and requests for refund. Commercial payment and dispute terms are confidential.

Chinese permits, customer orders and physical output remain separate revenue gates for the industry. The model assumes no automatic November 10 permit expiry or release.

4、合同期限：自合同生效之日起至商务部首次颁发及后续续期的两用物项和技术出口许可证载明的期限届满之日止；

5、生效条件：自双方签署之日起生效，合同履行以取得并维持两用物项出口许可为前提；

6、其他条款：如云南鑫耀未能按期取得或维持相关出口许可，AOI 有权取消合同或受影响的采购订单并要求退还相应款项。此外，合同对供应计划、付款约定、合同争议的解决等方面做出了明确约定。

根据交易双方签署的保密协议，本次签署协议中除已披露的数量、单价及商业金额外，部分详细定价、交付、付款及其他商业敏感条款信息属于商业秘密。根据《深圳证券交易所股票上市规则》《上市公司信息披露暂缓与豁免管理规定》以及公司内部相关规定，公司已履行内部信息披露豁免程序，豁免披露相关敏感合同条款。

四、合同对上市公司的影响

本次《长期供货协议》和《年度合同》的签订，有望为公司磷化铟产品带来潜在的订单需求，有利于拓宽公司的盈利渠道。若合同顺利履行，预计将对公司履约年度的经营成果产生积极影响，具体影响金额及影响的报告期将视出口许可、采购订单和合同履行的具体情况而定，以公司经审计确认的收入为准。

本次合同的签署，是子公司云南鑫耀与该客户在原有良好合作基础上的进一步深化合作，充分体现了客户对公司 4 英寸 N 型磷化铟衬底品质及批量供货能力的认可，有助于提升公司的市场声誉与综合竞争力。

本次《长期供货协议》和《年度合同》的履行不会对公司业务的独立性造成影响，公司的主要业务不会因履行该合同而对合同当事人形成依赖。

五、风险提示

1、本次签订的《长期供货协议》为双方基于长期合作而达成的框架性约定，其中每年 18 万片、五年预计 90 万片为 AOI 的预计采购安排，实际价格和交付数量以双方后续签订的年度合同、采购订单及商务部出口许可证批准数量为准。在后期协议履行实施过程中，存在因客户需求变化、政策及市场环境变化等因素，履行效果不达预期的风险。敬请广大投资者注意投资风险。

2、本次签订的《年度合同》履行需以取得并维持两用物项出口许可为前提，若许可审批进度或审批结果不及预期，或相关政策要求发生调整，导致公司未

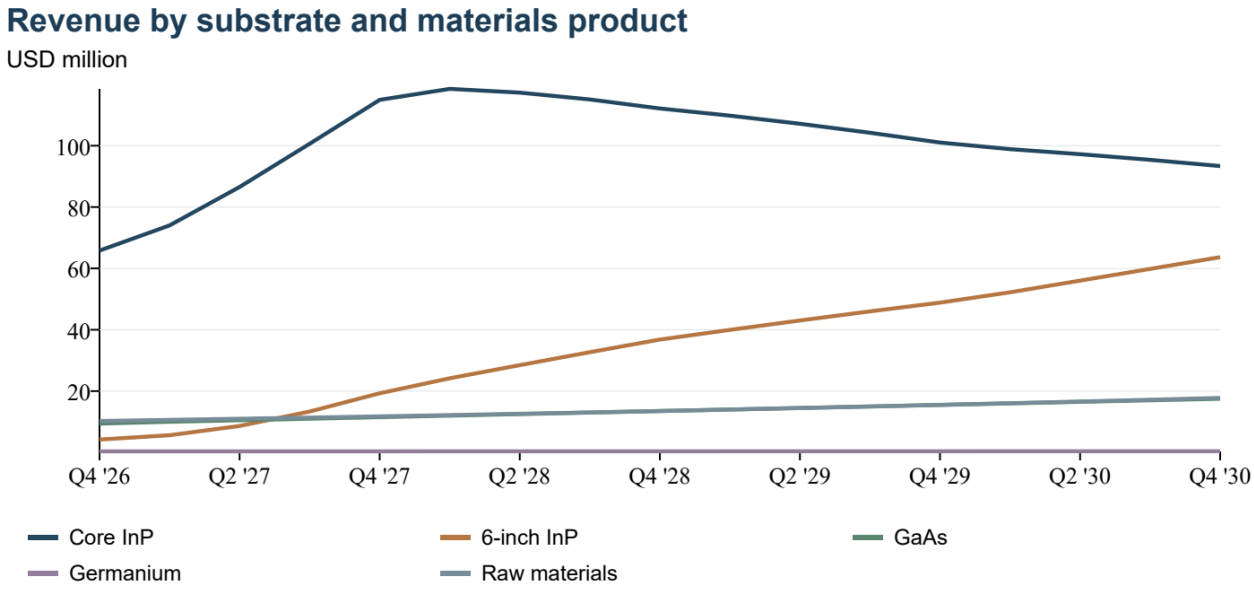
Yunnan Germanium announcement 2026-054, November 4, 2026, page 4. Original Mandarin terms describe export-permit conditions, cancellation/refund rights and confidential commercial terms.

15. Revenue mix and consolidation boundary

The five product vectors reconcile to consolidated external revenue. Internal material transfers are eliminated; unconsolidated affiliate sales are excluded.

GaAs starts at 9.4 and rises by 0.5 per modeled quarter after the first 2027 level of 10.0. Germanium follows the selected 0.35-0.39 path; raw-material revenue grows 3.5% per quarter from 10.25. Those trajectories are model assumptions.

Every future product point is an estimate. The numerical tables retain three decimals for tracing; the model keeps full precision.



Model estimates. Monetary amounts are USD millions unless specified.

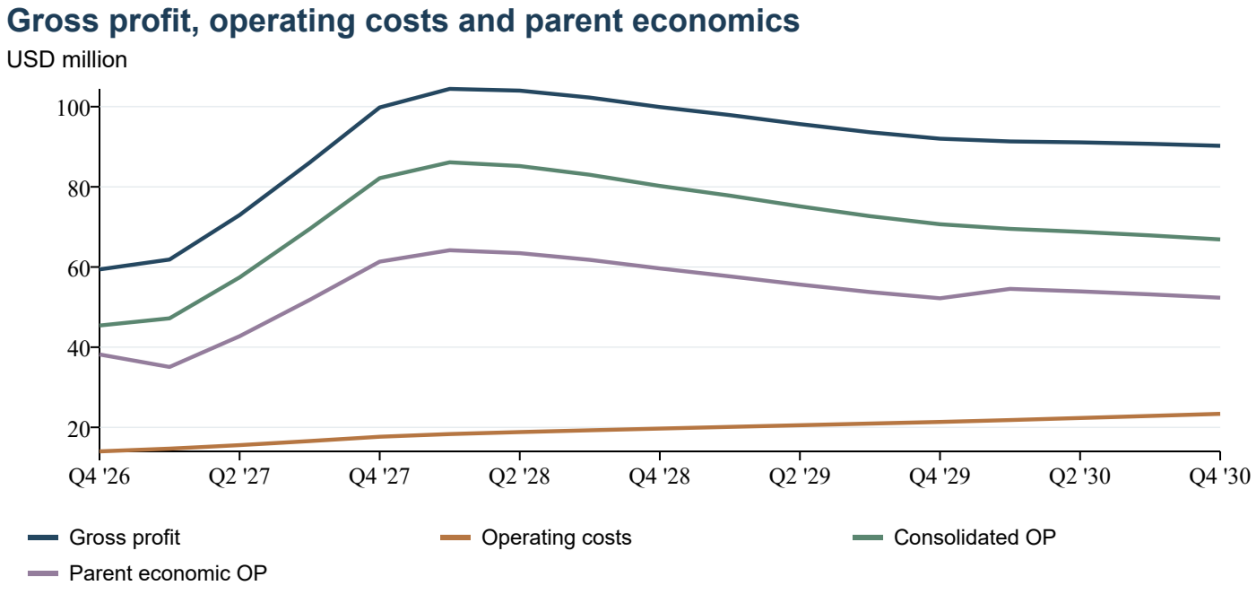
Quarter	Core InP	6-inch InP	GaAs	Germanium	Raw materials
Q4 2026	65.800	4.200	9.400	0.350	10.250
Q1 2027	74.008	5.613	10.000	0.360	10.609
Q2 2027	86.542	8.629	10.500	0.360	10.980
Q3 2027	100.627	13.333	11.000	0.360	11.364
Q4 2027	114.940	19.283	11.500	0.360	11.762
Q1 2028	118.527	24.171	12.000	0.370	12.174
Q2 2028	117.303	28.457	12.500	0.370	12.600
Q3 2028	115.077	32.680	13.000	0.370	13.041
Q4 2028	112.136	36.801	13.500	0.370	13.497
Q1 2029	109.797	39.949	14.000	0.380	13.970
Q2 2029	107.151	43.019	14.500	0.380	14.459
Q3 2029	104.221	45.989	15.000	0.380	14.965
Q4 2029	101.031	48.837	15.500	0.380	15.488
Q1 2030	98.877	52.205	16.000	0.390	16.031
Q2 2030	97.235	56.010	16.500	0.390	16.592
Q3 2030	95.393	59.835	17.000	0.390	17.172
Q4 2030	93.362	63.670	17.500	0.390	17.773

16. Production costs and operating leverage

Each product has its own gross-margin path. Group gross profit is the sum of product gross profits. Research and development and SG&A rise with time and incremental reference revenue.

R&D equals 4.35 times 1.025 to the quarter index plus 1.5% of reference revenue above 90. SG&A equals 9.65 times 1.02 to the quarter index plus 2.0% of the same incremental revenue. Existing policy support costs are additional and explicit.

Manufacturing depreciation and stock compensation remain inside the GAAP-style expense base. The cash-flow reconciliation adds those noncash amounts back once.



Model estimates. Monetary amounts are USD millions unless specified.

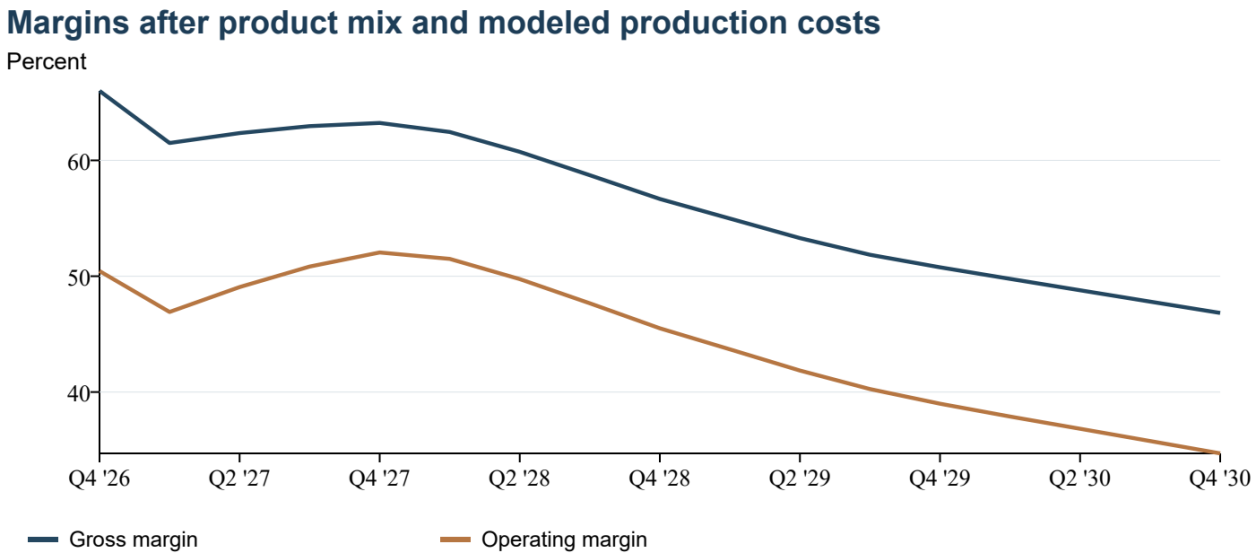
Quarter	Gross profit	Operating costs	Consolidated OP	Parent economic OP
Q4 2026	59.400	14.000	45.400	38.181
Q1 2027	61.863	14.672	47.190	35.059
Q2 2027	72.963	15.555	57.407	42.729
Q3 2027	86.050	16.570	69.480	51.792
Q4 2027	99.806	17.648	82.158	61.337
Q1 2028	104.443	18.322	86.121	64.184
Q2 2028	104.002	18.813	85.189	63.458
Q3 2028	102.272	19.260	83.011	61.783
Q4 2028	99.910	19.686	80.223	59.642
Q1 2029	97.913	20.108	77.805	57.664
Q2 2029	95.659	20.524	75.135	55.619
Q3 2029	93.619	20.935	72.684	53.748
Q4 2029	92.010	21.342	70.668	52.198
Q1 2030	91.331	21.813	69.518	54.540
Q2 2030	91.096	22.326	68.770	53.912
Q3 2030	90.729	22.842	67.887	53.175
Q4 2030	90.237	23.362	66.876	52.335

17. Margin path and sensitivity discipline

Core and six-inch margin trajectories differ. Later price normalization and competing supply can reduce margins even while accepted output grows.

Isolated ASP sensitivity keeps reference unit and fixed manufacturing costs unchanged. Volume sensitivity moves 80% of reference manufacturing cost with output and retains 20% as fixed. This cost split is a LOW-confidence diagnostic assumption.

Capex sensitivity changes depreciation against the selected reference cost base. The selected gross-margin path already contains baseline depreciation.



Consolidated gross and operating margins, percent. Model estimates.

Quarter	Gross margin	Operating margin
Q4 2026	66.000	50.444
Q1 2027	61.500	46.914
Q2 2027	62.355	49.061
Q3 2027	62.956	50.832
Q4 2027	63.231	52.050
Q1 2028	62.450	51.495
Q2 2028	60.738	49.751
Q3 2028	58.720	47.662
Q4 2028	56.669	45.503
Q1 2029	54.978	43.687
Q2 2029	53.289	41.856
Q3 2029	51.851	40.256
Q4 2029	50.768	38.992
Q1 2030	49.771	37.884
Q2 2030	48.786	36.829
Q3 2030	47.805	35.769
Q4 2030	46.829	34.705

18. Joint financing and operating probabilities

The six joint states sum to 100%: full/strong 45%, full/paced 15%, minimum/strong 10%, minimum/paced 20%, delay/paced 5% and delay/weak 5%. Thus full funding is 60%, minimum funding 30% and delayed funding 10%.

Definitive agreements with customary conditions support a 90% some-closing judgment. The operating mix is 55% strong,40% paced and 5% weak. Funding and operating execution are represented jointly.

Each closing state divides into 60% continuing equity/IPO and 40% redemption stress. Ten resulting leaves carry their own operating, cash, ownership and per-share calculations before probability weighting.

Leaf	Probability %	Funding	Operations	Exit treatment
full strong IPO	27.0	full	strong	IPO
full strong redemption	18.0	full	strong	redemption
full paced IPO	9.0	full	paced	IPO
full paced redemption	6.0	full	paced	redemption
minimum strong IPO	6.0	minimum	strong	IPO
minimum strong redemption	4.0	minimum	strong	redemption
minimum paced IPO	12.0	minimum	paced	IPO
minimum paced redemption	8.0	minimum	paced	redemption
delay paced	5.0	delay	paced	no_new_investors
delay weak	5.0	delay	weak	no_new_investors

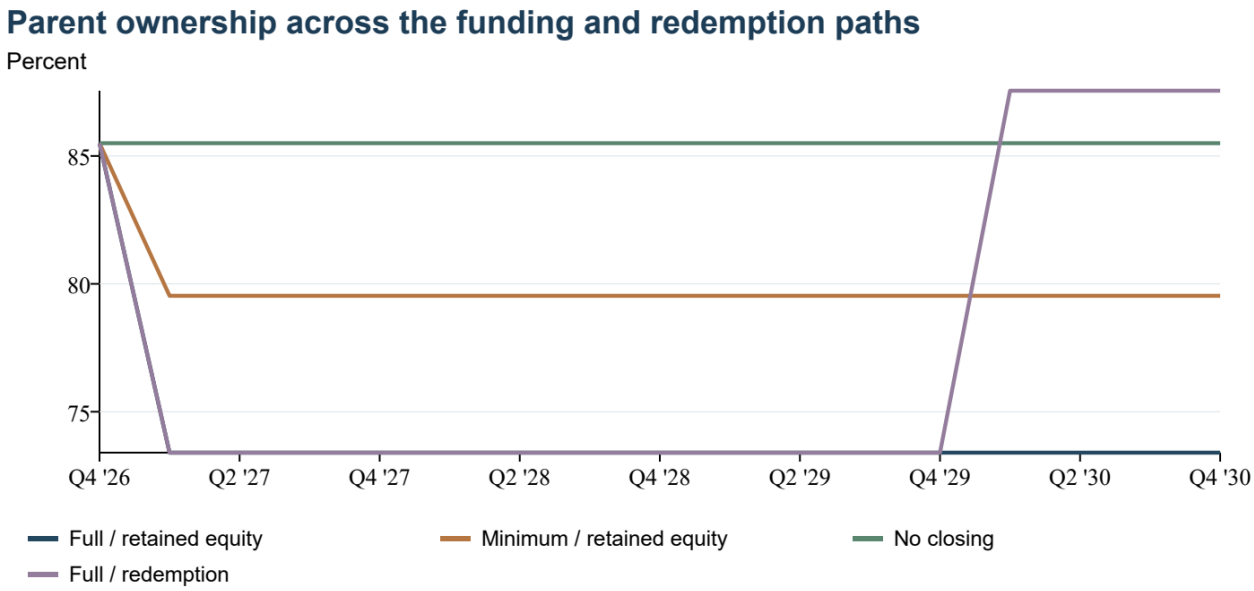
19. Tongmei ownership and external funding

October 27 agreements provide third-party investment of at leastRMB 1.91235 billion, approximatelyUSD 285 million, and up toRMB 3.668639 billion, approximatelyUSD 545 million. Expected closing isDecember 2026, subject to conditions.

AXT separately subscribesUSD 400 million equivalent for 342.998797 million Tongmei shares. The transfer moves funds inside the consolidated group. Only third-party proceeds increase group cash.

Maximum closing gives approximately 73.40% AXT ownership. The minimum-closing model uses 79.532993%, derived from the stated maximum outcome, prior 85.5%, parent subscription and a common issue-price proxy. The full minimum cap table remains unreported.

Financing closes at quarter-end in the model. Post-close ownership applies from Q1 2027. In the redemption leaves, cancellation restores the post-internal-recapitalization proxy of 87.551867% from Q1 2030.



AXT ownership of Tongmei, percent. Closing and redemption cases retain their stated assumptions.

Quarter	Full / retained equity	Minimum / retained equity	No closing	Full / redemption
Q4 2026	85.500	85.500	85.500	85.500
Q1 2027	73.400	79.533	85.500	73.400
Q2 2027	73.400	79.533	85.500	73.400
Q3 2027	73.400	79.533	85.500	73.400
Q4 2027	73.400	79.533	85.500	73.400
Q1 2028	73.400	79.533	85.500	73.400
Q2 2028	73.400	79.533	85.500	73.400
Q3 2028	73.400	79.533	85.500	73.400
Q4 2028	73.400	79.533	85.500	73.400
Q1 2029	73.400	79.533	85.500	73.400
Q2 2029	73.400	79.533	85.500	73.400
Q3 2029	73.400	79.533	85.500	73.400
Q4 2029	73.400	79.533	85.500	73.400
Q1 2030	73.400	79.533	85.500	87.552
Q2 2030	73.400	79.533	85.500	87.552
Q3 2030	73.400	79.533	85.500	87.552
Q4 2030	73.400	79.533	85.500	87.552

20. Parent economics and lower-tier minorities

The sanctioned parent operating-profit calculation is: (consolidated OP + parent-only costs) times Tongmei ownership, less parent-only costs and the lower-tier outside-owner claim.

Parent-only costs are 2.0 per quarter in 2026,2.5 in 2027,3.0 in 2028,3.5 in 2029 and 4.0 in 2030. These amounts are already included in consolidated operating costs.

The raw-material ownership proxy uses the legal chains and book-investment weights:3.758 at 100% local ownership,3.122 at 75%, and 7.331 at 58.5%. The resulting local raw ownership is 73.099254%. Book capital is a LOW-confidence proxy for the undisclosed earnings mix.

Raw OP equals raw GP less its revenue share of operating costs excluding parent costs. The outside-owner claim equals Tongmei ownership times 26.900746% times signed raw OP. The sign is preserved if that segment loses money.

21. Preference rights and the two redemption cohorts

The new investors have liquidation preference and priority on Tongmei profit distributions. The summary supplies no numerical distribution rate. A separate 10% parent-profit-allocation sensitivity illustrates this unresolved term.

The new ordinary no-IPO redemption trigger is the third closing anniversary. The stress branch pays original third-party principal compounded 8% for three years in Q4 2029. Earlier adverse-event triggers remain legal risks.

The 13% annual compounded return is part of the Qualified IPO definition. The 8% redemption accretion remains a contingent financing claim, outside recurring operating profit.

The model also pays a separate 49 legacy-investor redemption reserve in Q2 2027. Its possible ownership benefit remains unquantified, so the selected ownership path is retained.

AXT has an optional purchase mechanism using allowed forms of cash, note or common shares. The central redemption stress assumes Tongmei pays its obligation. A separate parent-liquidity stress examines resources without upstream dividends.

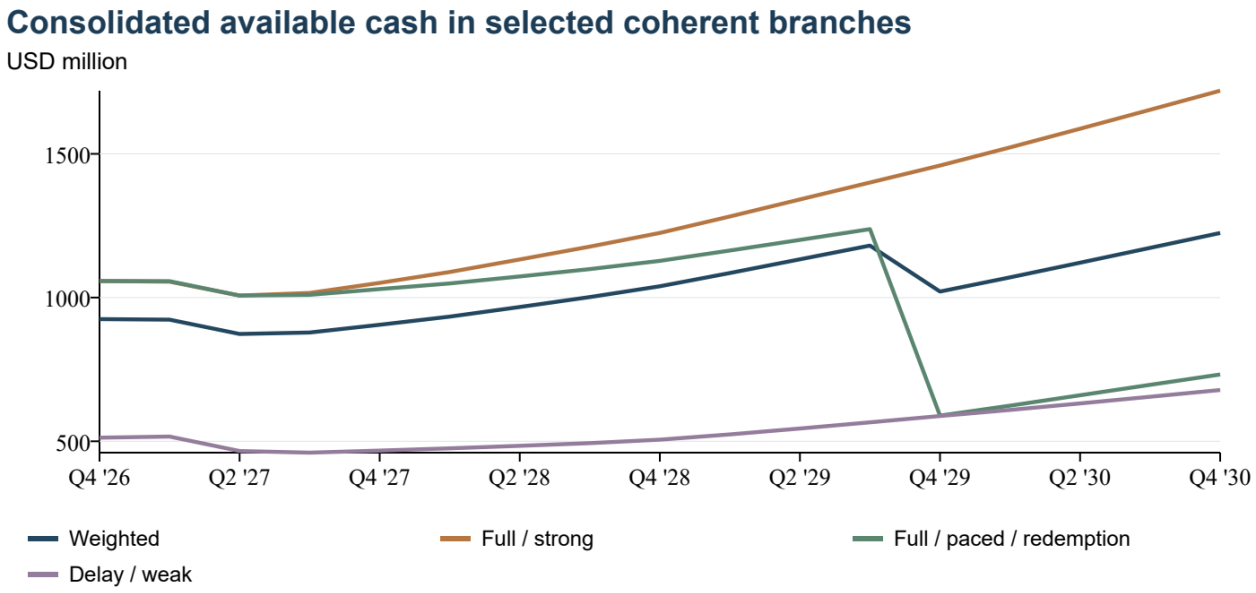
22. Consolidated cash and investment resources

September 30 cash is 485.472, short investments 27.266, long investments 271.956 and restricted cash 41.200. Short investments are converted at par into 512.738 initial available cash; long investments remain separate.

Interest uses 3% annual yield on opening available cash plus remaining long investments. Debt interest uses 5% of opening debt. Restricted cash stays outside spending resources.

Closing proceeds enter cash at quarter-end. The model first sells remaining long investments at par if required to maintain 25 cash. Any residual shortage raises common shares at 50% of the saved 88.58 reference price, with 5% issuance costs and next-quarter share dilution.

The selected group branches require no common issuance. Their minimum available cash is 453.961. These are conditional liquidity forecasts.



Model estimates. Monetary amounts are USD millions unless specified.

Quarter	Weighted	Full / strong	Full / paced / redemption	Delay / weak
Q4 2026	925.028	1,057.528	1,057.528	512.528
Q1 2027	923.109	1,055.300	1,056.987	516.416
Q2 2027	873.322	1,006.942	1,006.986	466.117
Q3 2027	878.356	1,016.056	1,009.451	460.417
Q4 2027	905.308	1,051.074	1,029.487	467.636
Q1 2028	933.678	1,088.915	1,049.129	475.422
Q2 2028	967.171	1,132.568	1,073.473	484.196
Q3 2028	1,001.712	1,177.208	1,099.029	493.478
Q4 2028	1,039.273	1,224.828	1,127.694	505.583
Q1 2029	1,085.388	1,282.136	1,163.660	523.878
Q2 2029	1,133.059	1,341.351	1,200.626	544.408
Q3 2029	1,180.813	1,400.288	1,238.073	565.910
Q4 2029	1,020.929	1,459.388	589.243	587.861
Q1 2030	1,070.271	1,522.476	623.952	609.045
Q2 2030	1,121.291	1,587.419	660.127	631.683
Q3 2030	1,172.928	1,653.358	696.408	654.889
Q4 2030	1,224.647	1,719.672	732.353	678.346

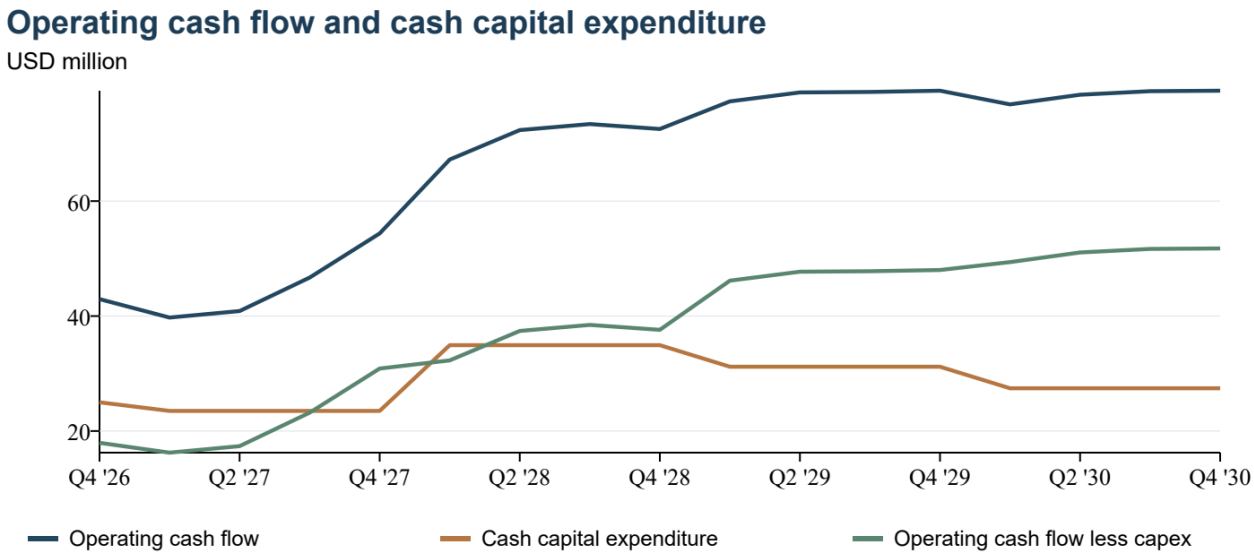
23. Operating cash flow and capital spending

Cash flow starts from consolidated income, adds back depreciation and stock compensation once, reconciles book versus cash tax, then incorporates receivables, inventory, other operating assets, payables and shipment credits.

AR uses 48 actual-quarter days of revenue; AP uses 50 days of COGS. Closing inventory is 70% of opening inventory plus 30% of 2.5 times quarterly COGS. Other operating assets move by 20% of the revenue change.

Strong annual cash capex is 100/160/140/120 in 2027-30. Paced is 90/120/110/100; weak is 60/75/75/75. Q4 2026 uses 25 in every branch. Annual amounts are split equally among their calendar quarters.

Conventional FCF equals operating cash flow less cash PP&E spending. Financing proceeds and investment liquidation appear separately.



Model estimates. Monetary amounts are USD millions unless specified.

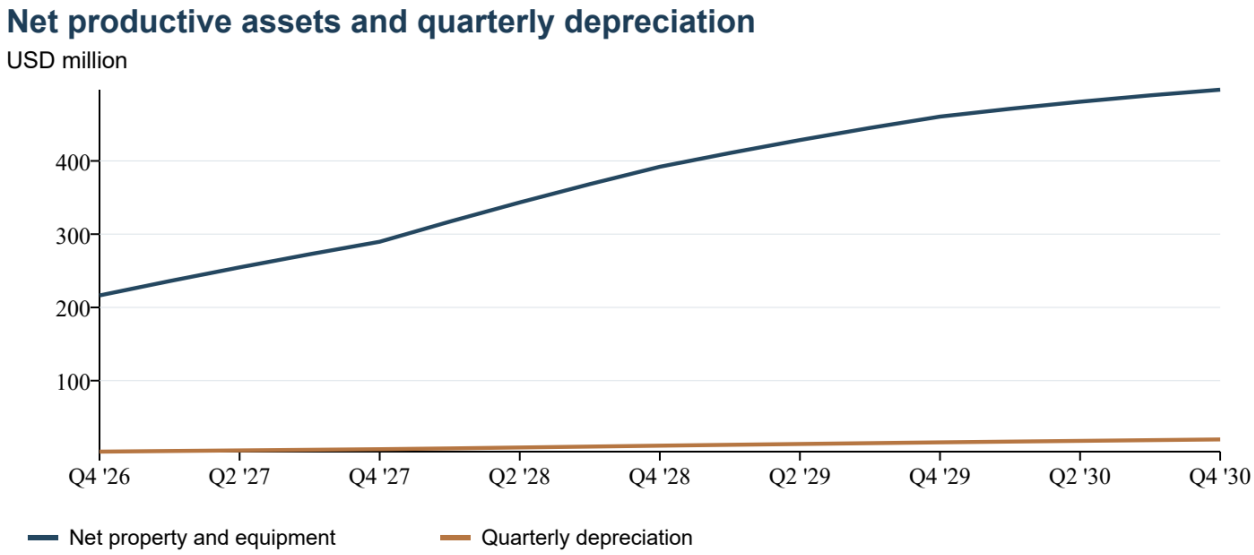
Quarter	Operating cash flow	Cash capital expenditure	Operating cash flow less capex
Q4 2026	42.950	25.000	17.950
Q1 2027	39.740	23.500	16.240
Q2 2027	40.872	23.500	17.372
Q3 2027	46.693	23.500	23.193
Q4 2027	54.371	23.500	30.871
Q1 2028	67.227	34.938	32.289
Q2 2028	72.349	34.938	37.412
Q3 2028	73.398	34.938	38.461
Q4 2028	72.542	34.938	37.605
Q1 2029	77.346	31.188	46.159
Q2 2029	78.902	31.188	47.715
Q3 2029	78.986	31.188	47.798
Q4 2029	79.200	31.188	48.013
Q1 2030	76.824	27.438	49.386
Q2 2030	78.502	27.438	51.064
Q3 2030	79.118	27.438	51.681
Q4 2030	79.201	27.438	51.763

24. Depreciation, assets and recurring costs

Legacy quarterly depreciation is 2.6, matching the reported Q3 call. New capex is depreciated over 28 quarters, with a half-quarter charge on addition and a full charge thereafter.

Net PP&E starts at 194.370 and rolls forward by capex less depreciation. The selected product cost base already includes reference depreciation; the ex-depreciation cost and depreciation amounts are kept separate.

Stock compensation is 1.1 times 1.04 to the quarter index, already expensed in the GAAP-style operating result. Its cash-flow addback has no second operating-profit benefit.



Model estimates. Monetary amounts are USD millions unless specified.

Quarter	Net property and equipment	Quarterly depreciation
Q4 2026	216.324	3.046
Q1 2027	235.911	3.913
Q2 2027	254.659	4.752
Q3 2027	272.568	5.591
Q4 2027	289.638	6.430
Q1 2028	317.101	7.474
Q2 2028	343.317	8.722
Q3 2028	368.285	9.969
Q4 2028	392.006	11.217
Q1 2029	410.795	12.398
Q2 2029	428.471	13.512
Q3 2029	445.033	14.626
Q4 2029	460.481	15.740
Q1 2030	471.132	16.786
Q2 2030	480.803	17.766
Q3 2030	489.494	18.746
Q4 2030	497.206	19.726

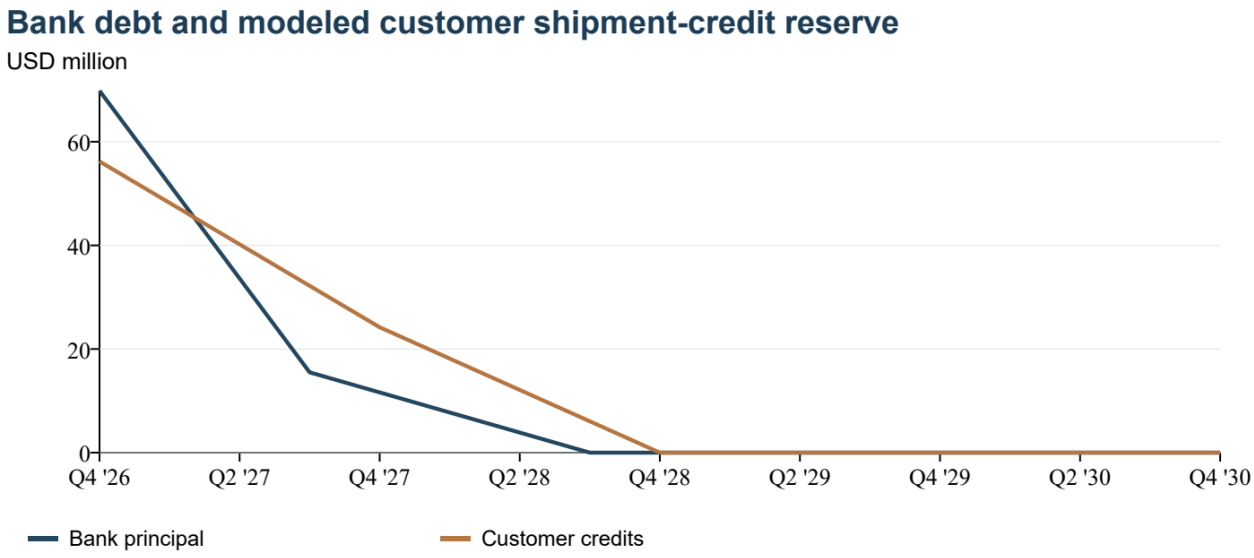
25. Customer credits and bank obligations

The modeled opening shipment-credit reserve is 56.2: the known gross receipt cohorts 12.7 plus Lumentum 43.5. The exact September 30 closing liability and Q3 shipment credits remain unavailable.

An additional Coherent 22.2885 receipt is unverified. The model assumes zero future deposit receipts and leaves that uncertainty visible.

Credits consume 8 per quarter during 2027 and 6.05 per quarter during 2028, capped by the remaining reserve. They reduce customer collections once while the product revenue follows delivery.

Short debt 72.461 is repaid in four equal installments from Q4 2026 through Q3 2027. The carried June long-debt estimate 15.5 is repaid from Q4 2027 through Q3 2028. Preferred cash dividends remain 0.044 per quarter.



Model estimates. Monetary amounts are USD millions unless specified.

Quarter	Bank principal	Customer credits
Q4 2026	69.846	56.200
Q1 2027	51.730	48.200
Q2 2027	33.615	40.200
Q3 2027	15.500	32.200
Q4 2027	11.625	24.200
Q1 2028	7.750	18.150
Q2 2028	3.875	12.100
Q3 2028	0.000	6.050
Q4 2028	0.000	0.000
Q1 2029	0.000	0.000
Q2 2029	0.000	0.000
Q3 2029	0.000	0.000
Q4 2029	0.000	0.000
Q1 2030	0.000	0.000
Q2 2030	0.000	0.000
Q3 2030	0.000	0.000
Q4 2030	0.000	0.000

26. Book tax, cash tax and the EPS diagnostic

Q4 book tax carries the observed Q3 effective rate 4.571/34.321, or 13.318377%. Cash tax is 15% of positive pretax income. Future book and cash tax both use 15%. The difference is explicitly reconciled in cash flow.

Q4 parent interest capture is 90% directly attributed to the parent plus 10% subject to Tongmei ownership. This LOW-confidence location proxy reflects dominant offshore investments and the dated June China liquidity disclosure.

Future net-interest capture uses Tongmei ownership as a conservative attribution proxy. No equity-method or other income is added to the forward earnings bridge.

At 66.1 million guidance shares, the model gives 0.563762 of Q4 GAAP-style common EPS. This is a consistency diagnostic for the selected operating and tax inputs. The operating-profit valuation retains the original 66.5/68/70/72/74 million annual economic share proxies.

27. Parent-only legal-liquidity stress

The separate no-upstream-dividend test begins with 90% of 512.738 liquid resources plus 271.956 long investments. It subtracts the 400 internal subscription on closing, parent overhead, preferred dividends and the 49 legacy reserve.

Parent investments earn the selected 3% opening yield. The stress applies 15% cash tax to positive parent interest less parent overhead and sells long investments at par to maintain 25 cash.

Across the closing paths, maximum cumulative long-investment realization is 27.759985 and no additional funding is required. These assets overlap the group resources and are excluded from any additive valuation calculation.

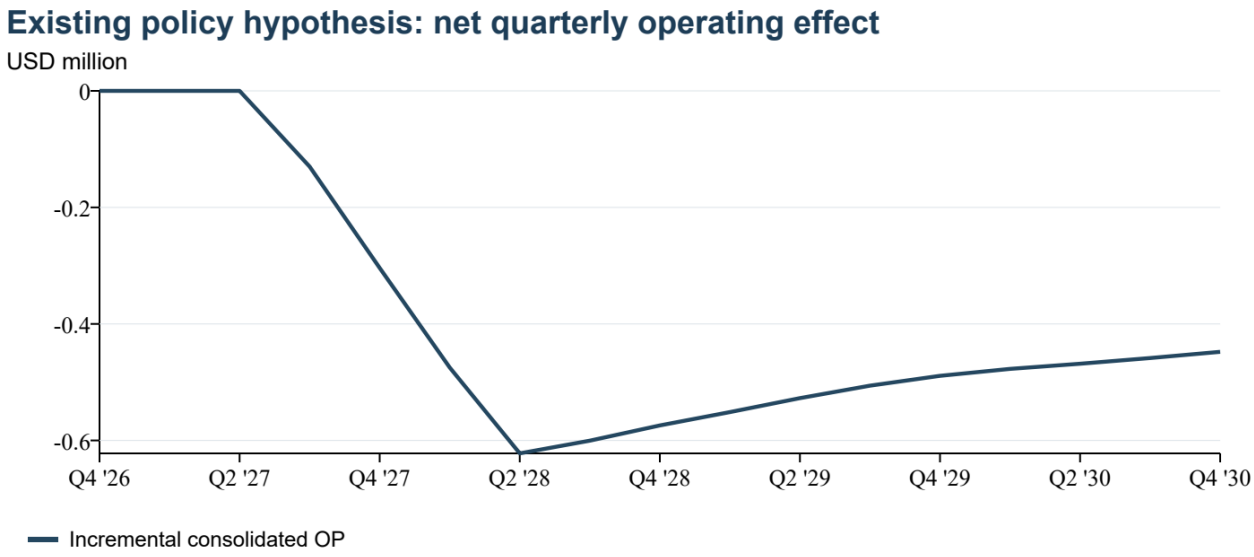
Actual entity-level cash, transfer restrictions, priority distributions and settlement obligations require the new filing and complete legal documents.

28. Existing policy hypothesis and overlap

The original FCC-optics hypothesis retains 40% common probability. Conditional on that event,75% allows a 5% price premium on 20% uncommitted InP sales;25% applies a 10% InP volume loss with 10% of the displaced COGS stranded.

Timing is zero through Q2 2027,25% in Q3 2027,50% in Q4 2027,75% in Q1 2028 and 100% from Q2 2028. Added manufacturing and operating support each consume 5% of the allowed-route incremental price revenue.

The overlay applies once to each coherent operating branch. Existing Chinese permit risk and the new backlog evidence receive no duplicated policy revenue. A November 10 automatic permit release remains unverified.



Model estimates. Monetary amounts are USD millions unless specified.

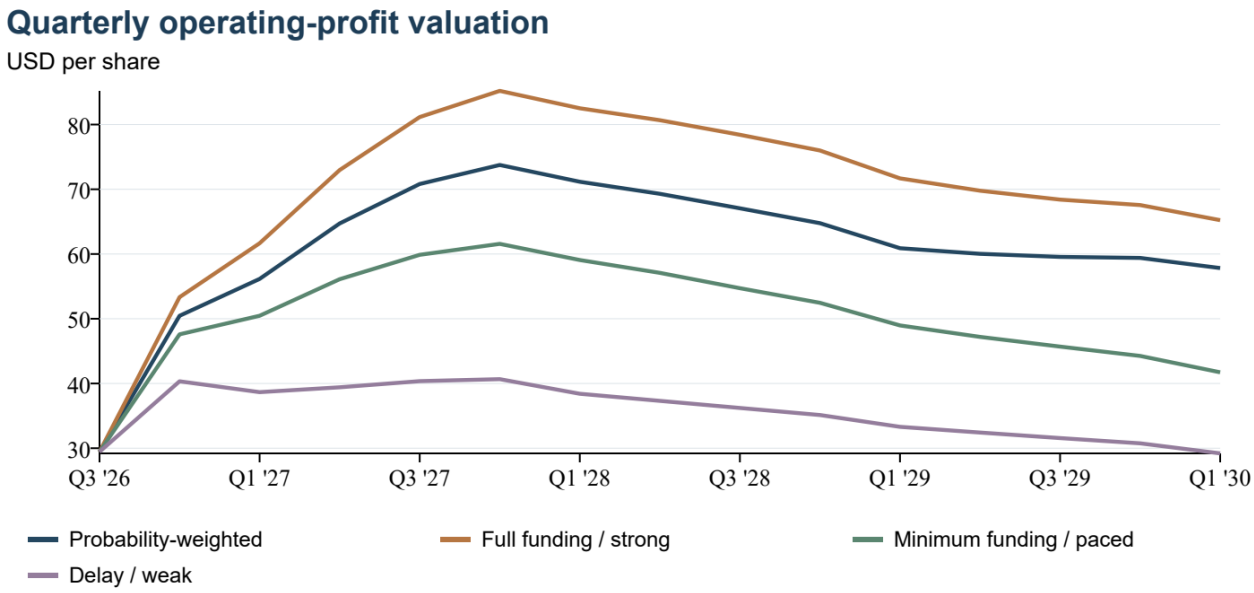
Quarter	Incremental consolidated OP
Q4 2026	0.000
Q1 2027	0.000
Q2 2027	0.000
Q3 2027	-0.130
Q4 2027	-0.304
Q1 2028	-0.475
Q2 2028	-0.622
Q3 2028	-0.600
Q4 2028	-0.574
Q1 2029	-0.551
Q2 2029	-0.527
Q3 2029	-0.506
Q4 2029	-0.489
Q1 2030	-0.477
Q2 2030	-0.468
Q3 2030	-0.459
Q4 2030	-0.448

29. Quarterly valuation path

Every future ceiling equals the branch's four-quarter parent-economic OP times 20 divided by that ceiling quarter's economic diluted-share proxy. Branch per-share values are then probability weighted.

Cash balances, prospective IPO market values and analyst targets supply no additional valuation term. Funding affects operating feasibility, ownership, redemption cash and any required common-share denominator.

The Q3 2026 published forecast retains its original 29.441593 value and information vintage. The chart shows that frozen historical point separately from the current and future forecast path.



USD per share. Q3 2026 preserves its published vintage; subsequent periods use the selected operating model.

Quarter	Probability-weighted	Full funding / strong	Minimum funding / paced	Delay / weak
Q3 2026	29.442	29.442	29.442	29.442
Q4 2026	50.455	53.332	47.591	40.334
Q1 2027	56.152	61.660	50.459	38.660
Q2 2027	64.718	72.968	56.107	39.410
Q3 2027	70.815	81.155	59.874	40.359
Q4 2027	73.753	85.201	61.567	40.668
Q1 2028	71.162	82.515	59.071	38.410
Q2 2028	69.299	80.669	57.088	37.315
Q3 2028	67.059	78.421	54.718	36.217
Q4 2028	64.764	75.992	52.462	35.127
Q1 2029	60.897	71.685	48.966	33.303
Q2 2029	60.029	69.780	47.197	32.420
Q3 2029	59.555	68.403	45.697	31.564
Q4 2029	59.396	67.562	44.256	30.755
Q1 2030	57.828	65.228	41.726	29.203

30. Exact forward-four-quarter derivation

Each window below uses the ceiling quarter and the following three quarters. Values are displayed to six decimals here; calculations retain full precision.

Future final prices are weighted after each leaf uses its own operating profit, ownership and share count. Weighted OP and weighted shares are descriptive diagnostics.

Ceiling quarter	Earnings window	F4Q parent OP	Multiple	Shares M	Ceiling USD
Q3 2026	Q3 2026 to Q2 2027	97.893297	20.0	66.500	29.441593
Q4 2026	Q4 2026 to Q3 2027	167.761479	20.0	66.500	50.454580
Q1 2027	Q1 2027 to Q4 2027	190.917016	20.0	68.000	56.152064
Q2 2027	Q2 2027 to Q1 2028	220.042267	20.0	68.000	64.718314
Q3 2027	Q3 2027 to Q2 2028	240.770752	20.0	68.000	70.814927
Q4 2027	Q4 2027 to Q3 2028	250.761290	20.0	68.000	73.753321
Q1 2028	Q1 2028 to Q4 2028	249.067010	20.0	70.000	71.162003
Q2 2028	Q2 2028 to Q1 2029	242.546430	20.0	70.000	69.298980
Q3 2028	Q3 2028 to Q2 2029	234.707315	20.0	70.000	67.059233
Q4 2028	Q4 2028 to Q3 2029	226.672564	20.0	70.000	64.763590
Q1 2029	Q1 2029 to Q4 2029	219.228036	20.0	72.000	60.896677
Q2 2029	Q2 2029 to Q1 2030	216.104976	20.0	72.000	60.029160
Q3 2029	Q3 2029 to Q2 2030	214.398583	20.0	72.000	59.555162
Q4 2029	Q4 2029 to Q3 2030	213.826137	20.0	72.000	59.396149
Q1 2030	Q1 2030 to Q4 2030	213.963400	20.0	74.000	57.827946

31. Revenue and gross-profit reconciliation

All values are probability-weighted USD millions. Each product table maps directly to the model product vector.

Quarter	Revenue	Gross profit	COGS	R&D	SG&A	Policy opex
Q4 2026	90.000	59.400	30.600	4.350	9.650	0.000
Q1 2027	100.590	61.863	38.727	4.618	10.055	0.000
Q2 2027	117.011	72.963	44.048	4.975	10.580	0.000
Q3 2027	136.684	86.050	50.634	5.388	11.178	0.004
Q4 2027	157.845	99.806	58.039	5.826	11.812	0.010
Q1 2028	167.242	104.443	62.799	6.092	12.214	0.016
Q2 2028	171.231	104.002	67.228	6.279	12.513	0.022
Q3 2028	174.168	102.272	71.897	6.449	12.789	0.022
Q4 2028	176.305	99.910	76.395	6.610	13.054	0.022
Q1 2029	178.095	97.913	80.182	6.770	13.316	0.023
Q2 2029	179.508	95.659	83.850	6.927	13.575	0.023
Q3 2029	180.555	93.619	86.936	7.082	13.831	0.023
Q4 2029	181.236	92.010	89.226	7.235	14.084	0.023
Q1 2030	183.502	91.331	92.171	7.415	14.375	0.023
Q2 2030	186.726	91.096	95.631	7.614	14.689	0.023
Q3 2030	189.790	90.729	99.061	7.813	15.005	0.023
Q4 2030	192.695	90.237	102.458	8.015	15.323	0.024

32. Parent-profit reconciliation

Parent costs are already within operating costs. The outside-owner adjustment is applied once.

Displayed OP and ownership are weighted diagnostics. Each branch's parent-economic OP is calculated first, then weighted. Multiplying the two displayed weighted diagnostics would discard the relationship between funding and operations.

Quarter	Group OP	Parent cost	Ownership %	Raw claim	Parent OP
Q4 2026	45.400	2.000	85.500	0.346	38.181
Q1 2027	47.190	2.500	76.450	0.346	35.059
Q2 2027	57.407	2.500	76.450	0.379	42.729
Q3 2027	69.480	2.500	76.450	0.411	51.792
Q4 2027	82.158	2.500	76.450	0.442	61.337
Q1 2028	86.121	3.000	76.450	0.468	64.184
Q2 2028	85.189	3.000	76.450	0.483	63.458
Q3 2028	83.011	3.000	76.450	0.497	61.783
Q4 2028	80.223	3.000	76.450	0.511	59.642
Q1 2029	77.805	3.500	76.450	0.533	57.664
Q2 2029	75.135	3.500	76.450	0.547	55.619
Q3 2029	72.684	3.500	76.450	0.560	53.748
Q4 2029	70.668	3.500	76.450	0.574	52.198
Q1 2030	69.518	4.000	80.809	0.633	54.540
Q2 2030	68.770	4.000	80.809	0.651	53.912
Q3 2030	67.887	4.000	80.809	0.669	53.175
Q4 2030	66.876	4.000	80.809	0.687	52.335

33a. Core InP product path

The product allocation, margin and gross profit are model estimates. The policy effect is already included where applicable.

Quarter	Revenue	Gross margin %	Gross profit	COGS
Q4 2026	65.800	78.016	51.335	14.465
Q1 2027	74.008	71.316	52.780	21.228
Q2 2027	86.542	71.556	61.925	24.616
Q3 2027	100.627	71.469	71.918	28.709
Q4 2027	114.940	71.108	81.732	33.208
Q1 2028	118.527	69.949	82.908	35.619
Q2 2028	117.303	67.927	79.681	37.622
Q3 2028	115.077	65.902	75.839	39.239
Q4 2028	112.136	63.833	71.579	40.557
Q1 2029	109.797	61.864	67.924	41.873
Q2 2029	107.151	59.897	64.180	42.971
Q3 2029	104.221	58.296	60.757	43.464
Q4 2029	101.031	57.288	57.879	43.152
Q1 2030	98.877	56.278	55.645	43.231
Q2 2030	97.235	55.287	53.759	43.476
Q3 2030	95.393	54.298	51.796	43.597
Q4 2030	93.362	53.309	49.770	43.592

33b. Six-inch InP product path

The product allocation, margin and gross profit are model estimates. The policy effect is already included where applicable.

Quarter	Revenue	Gross margin %	Gross profit	COGS
Q4 2026	4.200	50.000	2.100	2.100
Q1 2027	5.613	49.393	2.772	2.840
Q2 2027	8.629	51.689	4.460	4.169
Q3 2027	13.333	54.645	7.286	6.047
Q4 2027	19.283	56.806	10.954	8.329
Q1 2028	24.171	58.226	14.074	10.097
Q2 2028	28.457	58.249	16.576	11.881
Q3 2028	32.680	56.300	18.399	14.281
Q4 2028	36.801	54.355	20.003	16.798
Q1 2029	39.949	53.299	21.292	18.657
Q2 2029	43.019	52.252	22.478	20.541
Q3 2029	45.989	51.213	23.552	22.437
Q4 2029	48.837	50.181	24.507	24.330
Q1 2030	52.205	49.151	25.659	26.546
Q2 2030	56.010	48.177	26.984	29.026
Q3 2030	59.835	47.208	28.247	31.588
Q4 2030	63.670	46.244	29.443	34.227

33c. GaAs product path

The product allocation, margin and gross profit are model estimates. The policy effect is already included where applicable.

Quarter	Revenue	Gross margin %	Gross profit	COGS
Q4 2026	9.400	32.000	3.008	6.392
Q1 2027	10.000	32.500	3.250	6.750
Q2 2027	10.500	32.500	3.412	7.088
Q3 2027	11.000	32.500	3.575	7.425
Q4 2027	11.500	32.500	3.738	7.763
Q1 2028	12.000	33.000	3.960	8.040
Q2 2028	12.500	33.000	4.125	8.375
Q3 2028	13.000	33.000	4.290	8.710
Q4 2028	13.500	33.000	4.455	9.045
Q1 2029	14.000	33.500	4.690	9.310
Q2 2029	14.500	33.500	4.857	9.643
Q3 2029	15.000	33.500	5.025	9.975
Q4 2029	15.500	33.500	5.192	10.308
Q1 2030	16.000	34.000	5.440	10.560
Q2 2030	16.500	34.000	5.610	10.890
Q3 2030	17.000	34.000	5.780	11.220
Q4 2030	17.500	34.000	5.950	11.550

33d. Germanium product path

The product allocation, margin and gross profit are model estimates. The policy effect is already included where applicable.

Quarter	Revenue	Gross margin %	Gross profit	COGS
Q4 2026	0.350	25.000	0.088	0.262
Q1 2027	0.360	25.000	0.090	0.270
Q2 2027	0.360	25.000	0.090	0.270
Q3 2027	0.360	25.000	0.090	0.270
Q4 2027	0.360	25.000	0.090	0.270
Q1 2028	0.370	25.000	0.093	0.277
Q2 2028	0.370	25.000	0.093	0.277
Q3 2028	0.370	25.000	0.093	0.277
Q4 2028	0.370	25.000	0.093	0.277
Q1 2029	0.380	25.000	0.095	0.285
Q2 2029	0.380	25.000	0.095	0.285
Q3 2029	0.380	25.000	0.095	0.285
Q4 2029	0.380	25.000	0.095	0.285
Q1 2030	0.390	25.000	0.098	0.292
Q2 2030	0.390	25.000	0.098	0.292
Q3 2030	0.390	25.000	0.098	0.292
Q4 2030	0.390	25.000	0.098	0.292

33e. Raw materials product path

The product allocation, margin and gross profit are model estimates. The policy effect is already included where applicable.

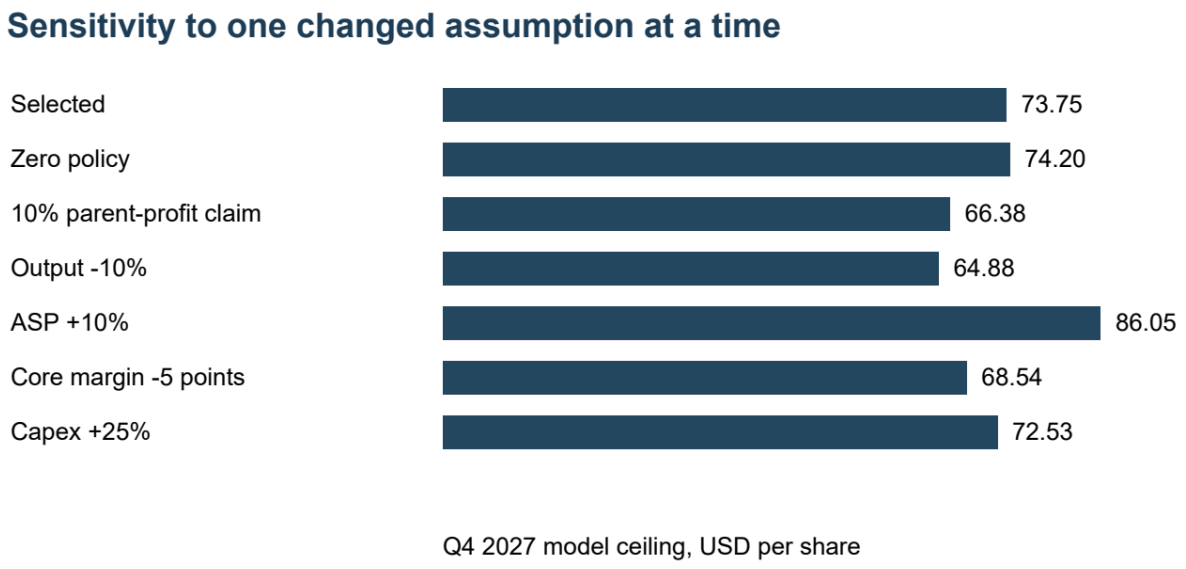
Quarter	Revenue	Gross margin %	Gross profit	COGS
Q4 2026	10.250	28.000	2.870	7.380
Q1 2027	10.609	28.000	2.970	7.638
Q2 2027	10.980	28.000	3.074	7.906
Q3 2027	11.364	28.000	3.182	8.182
Q4 2027	11.762	28.000	3.293	8.469
Q1 2028	12.174	28.000	3.409	8.765
Q2 2028	12.600	28.000	3.528	9.072
Q3 2028	13.041	28.000	3.651	9.389
Q4 2028	13.497	28.000	3.779	9.718
Q1 2029	13.970	28.000	3.912	10.058
Q2 2029	14.459	28.000	4.048	10.410
Q3 2029	14.965	28.000	4.190	10.775
Q4 2029	15.488	28.000	4.337	11.152
Q1 2030	16.031	28.000	4.489	11.542
Q2 2030	16.592	28.000	4.646	11.946
Q3 2030	17.172	28.000	4.808	12.364
Q4 2030	17.773	28.000	4.977	12.797

34. Assumption sensitivity

Each sensitivity changes one stated input while retaining the other selected assumptions and the 20x operating-profit method. Output, ASP, manufacturing cost, depreciation and operating support recalculate together.

The 10% parent-profit allocation case illustrates the unresolved preference term. The zero-policy case removes only the existing incremental policy hypothesis.

These values describe conditional model exposure. They carry no additional probability beyond the selected branch calculation.



Q4 2027 model ceiling in USD per share, changing one stated assumption at a time.

Changed assumption	Q4 2027 ceiling USD
Selected	73.753
Zero policy	74.198
10% parent-profit claim	66.378
Output -10%	64.876
ASP +10%	86.050
Core margin -5 points	68.544
Capex +25%	72.535

35. Numerical operating probability ledger

Q4 2027 decomposition of the selected case. The 55% strong, 40% paced and 5% weak weights arise from the joint funding and operating states. Conditional parent earnings also reflect the financing mix inside each operating group.

Sources: October 29 management output, allocation and backlog discussion (AXTI-F11 to F 13), plus the November 4 Yunnan/AOI contract (AXTI-NOV4-AOI-RIVAL). Reference Q4 2026 InP revenue is 70. Existing contracts and backlog already sit inside the output paths.

Weighted contributions sum to the selected InP revenue, consolidated OP and parent OP. This table decomposes the current case and introduces no additional revenue or probability multiplier.

Outcome	p %	Conditional InP	Conditional OP	Weighted InP	Weighted OP	Weighted parent OP
strong	55.0	152.345	97.550	83.790	53.653	39.382
paced	40.0	115.096	65.956	46.038	26.382	20.179
weak	5.0	87.891	42.462	4.395	2.123	1.776
Weighted total	100.0	-	-	134.223	82.158	61.337

35b. Financing and overlap ledger

Sources: October 28 issuer financing release and 8-K (AXTI-F16 to F 20). Weighted external proceeds are 412.5 and affect cash only. Weighted Q1 2027 ownership is 76.450%. The internal parent subscription has an expected 360 transfer and zero consolidated cash increment.

The existing October 1 policy hypothesis carries 40% probability once. Its Q4 2027 forward-window parent-OP contribution is -1.512782 million, corresponding to -0.444936 per share under the same 20x and 68 million shares. The zero-policy control supplies this isolated difference.

The October 29 backlog above 500 and the November 4 relay have zero separate revenue addition: signed, short-duration and spot demand supports the selected output distribution. The 194.4 rival contract also has zero direct subtraction from AXT sales: it supports the competitive outcomes without identifying displaced AXT invoices.

Update tests: accepted wafer output by grade, customer drawdowns, like-for-like pricing, competitor shipment acceptance, financing closings, full preference terms and permit conversion.

Funding	p %	External cash	Weighted cash	Postclose own %	Weighted own %
full	60.0	545.000	327.000	73.400	44.040
minimum	30.0	285.000	85.500	79.533	23.860
delay	10.0	0.000	0.000	85.500	8.550

36. Cross-book interpretation and emerging competition

Lumentum and Coherent are known customer counterparties. Their optical-device expansion can support merchant substrate purchases, while integration and alternative qualification can change the size of that wallet.

AAOI's disclosed Yunnan relationship provides fresh evidence that another four-inch source can serve a relevant downstream buyer. The contract is separate from AXT's existing orders and deserves continued tracking.

The technical expert discussion identifies substitution and productivity risks across emitter technologies and wafer sizes. These risks belong to product- and customer-specific estimates.

37. Dated catalysts and adverse tests

December 2026: verify minimum/maximum financing closing, actual ownership, currency transfer, registration and legal preferences.

Q4 results and early 2027: test approximately 90 revenue,66% modeled gross margin, operating-cost absorption, customer credits and permit-covered shipments.

2027: compare physical expansion with accepted output, confirm six-inch and four-inch qualification, and track contract fulfillment and competing-source deliveries.

2028: check whether greater supply and customer inventories reduce comparable pricing and gross margins.

Q4 2029 onward: monitor qualified-IPO progress and redemption terms. A triggered obligation can consume cash even when the operating business remains profitable.

38. Source completeness and model limitations

The latest Q3 earnings release is retained; the latest full 10-Q in the verified source set is for June 30, filed August 13. Q3 operating cash flow, detailed liabilities, legal common shares, tax and subsidiary allocations remain unavailable.

The latest two full displayed management calls were cross-compared through closing. A host labels its access as preview despite the visible complete sequence. The original issuer audio remains unrecovered.

The full October 28 financing 8-K summary is available. Complete subscription, investor-rights and support agreements were indicated for the 2026 10-K. Preference and minimum-close ownership details remain bounded by the available summary.

Exact wafer-equivalent output, accepted product grades, realized ASP, diameter margins and customer buffer inventory remain unreported. The model uses explicitly labeled output, pricing, cost and ownership proxies.

The report retains each limitation beside the affected analysis. Future results and legal disclosures can change the probability, timing, cash and per-share outcome.

39. Source register

Primary source dates, source-chain overlap and the retained Mandarin notices govern the numerical interpretation. The private model retains the source identifiers, selected inputs and full precision.

Source	Date / scope	Reference
AXT Q3 release	Oct29,2026; Sep30 quarter	https://investors.axt.com/Investors/news/news-details/2026/AXT-Inc--Announces-Third-Quarter-2026-Financial-Results/default.aspx
Q3 management call	Oct29,2026; visible full Q&A	https://earningscalls.dev/transcripts/axt-inc_axti_earnings_call_transcript_2026-10-29
Q2 release and call	Jul30,2026; June30 quarter	https://earningscalls.dev/transcripts/axt-inc_axti_earnings_call_transcript_2026-07-30
Q2 10-Q	Aug13,2026	https://www.sec.gov/Archives/edgar/data/1051627/000143774926027677/axti20260630_10q.htm
Tongmei financing8-K	Oct28,2026; Oct27 agreements	https://www.sec.gov/Archives/edgar/data/1051627/000121390026113058/ea0309774-8k_axt.htm
Yunnan/AOI contract	Nov4,2026; announcement2026-054	https://static.cninfo.com.cn/finalpage/2026-11-04/1225655614.PDF
P Equity InP expert call	PublishedOct19,2026; interview day unknown	Retained original pages7 and9
Serenity policy chain	Oct1,2026	https://x.com/aleabitoreddit/status/2105709782854435274 ; underlying full MS report unavailable
China export-control regime	Official policy source	https://policy.mofcom.gov.cn/claw/clawContent.shtml?id=101925

40. Historical forecast and current reconstruction

The published Q3 2026 forecast is preserved at its original 66.000 revenue,29.8708 GP,18.8708 group OP,15.844534 parent OP and 29.441593 ceiling. Its original four-quarter window and conventions remain intact.

Reported Q3 actuals appear in the separate latest-quarter section. Current and future forecasts use those results as their calibration evidence. The historical display preserves the information that was available when it was published.

Original historical formulas and source files remain in recoverable snapshots. The current model's historical entry and the workbook display preserve that vintage.

Appendix A. full strong IPO

Probability 27.0%. Funding full; operations strong; exit IPO. All values are USD millions except ownership and shares.

Each leaf calculates operating results, lower-tier claims, cash resources and ownership before any portfolio-weighted per-share calculation.

Quarter	Revenue	GP	Opex	Group OP	Parent OP	Own %	Shares M
Q4 2026	90.000	59.400	14.000	45.400	38.181	85.500	66.500
Q1 2027	103.289	64.642	14.767	49.876	35.606	73.400	68.000
Q2 2027	123.900	79.244	15.797	63.447	45.531	73.400	68.000
Q3 2027	149.552	97.523	17.022	80.501	58.012	73.400	68.000
Q4 2027	175.967	115.836	18.286	97.550	70.495	73.400	68.000
Q1 2028	185.395	121.573	18.963	102.610	74.052	73.400	70.000
Q2 2028	189.687	121.166	19.467	101.699	73.369	73.400	70.000
Q3 2028	193.235	119.475	19.935	99.540	71.769	73.400	70.000
Q4 2028	195.755	116.998	20.375	96.623	69.614	73.400	70.000
Q1 2029	197.815	114.884	20.806	94.078	67.591	73.400	72.000
Q2 2029	199.464	112.478	21.231	91.247	65.499	73.400	72.000
Q3 2029	200.714	109.875	21.649	88.226	63.267	73.400	72.000
Q4 2029	201.564	108.184	22.061	86.123	61.708	73.400	72.000
Q1 2030	204.145	107.552	22.543	85.009	60.734	73.400	74.000
Q2 2030	208.647	107.875	23.102	84.773	60.542	73.400	74.000
Q3 2030	213.029	108.051	23.665	84.386	60.239	73.400	74.000
Q4 2030	217.293	108.085	24.232	83.852	59.828	73.400	74.000

Appendix B. full strong IPO cash

Closing bank balances reconcile beginning cash, operating cash flow, cash capex, external financing, bank repayment, redemption, preferred dividends, investment realization and any common issuance.

Quarter	CFO	Capex	External PE	Debt repaid	Redemption	LT sold	End cash
Q4 2026	42.950	25.000	545.000	18.115	0.000	0.000	1,057.528
Q1 2027	40.930	25.000	0.000	18.115	0.000	0.000	1,055.300
Q2 2027	43.801	25.000	0.000	18.115	49.000	0.000	1,006.942
Q3 2027	52.273	25.000	0.000	18.115	0.000	0.000	1,016.056
Q4 2027	63.938	25.000	0.000	3.875	0.000	0.000	1,051.074
Q1 2028	81.759	40.000	0.000	3.875	0.000	0.000	1,088.915
Q2 2028	87.572	40.000	0.000	3.875	0.000	0.000	1,132.568
Q3 2028	88.559	40.000	0.000	3.875	0.000	0.000	1,177.208
Q4 2028	87.664	40.000	0.000	0.000	0.000	0.000	1,224.828
Q1 2029	92.352	35.000	0.000	0.000	0.000	0.000	1,282.136
Q2 2029	94.259	35.000	0.000	0.000	0.000	0.000	1,341.351
Q3 2029	93.980	35.000	0.000	0.000	0.000	0.000	1,400.288
Q4 2029	94.145	35.000	0.000	0.000	0.000	0.000	1,459.388
Q1 2030	93.132	30.000	0.000	0.000	0.000	0.000	1,522.476
Q2 2030	94.987	30.000	0.000	0.000	0.000	0.000	1,587.419
Q3 2030	95.983	30.000	0.000	0.000	0.000	0.000	1,653.358
Q4 2030	96.358	30.000	0.000	0.000	0.000	0.000	1,719.672

Appendix A. full strong redemption

Probability 18.0%. Funding full; operations strong; exit redemption. All values are USD millions except ownership and shares.

Each leaf calculates operating results, lower-tier claims, cash resources and ownership before any portfolio-weighted per-share calculation.

Quarter	Revenue	GP	Opex	Group OP	Parent OP	Own %	Shares M
Q4 2026	90.000	59.400	14.000	45.400	38.181	85.500	66.500
Q1 2027	103.289	64.642	14.767	49.876	35.606	73.400	68.000
Q2 2027	123.900	79.244	15.797	63.447	45.531	73.400	68.000
Q3 2027	149.552	97.523	17.022	80.501	58.012	73.400	68.000
Q4 2027	175.967	115.836	18.286	97.550	70.495	73.400	68.000
Q1 2028	185.395	121.573	18.963	102.610	74.052	73.400	70.000
Q2 2028	189.687	121.166	19.467	101.699	73.369	73.400	70.000
Q3 2028	193.235	119.475	19.935	99.540	71.769	73.400	70.000
Q4 2028	195.755	116.998	20.375	96.623	69.614	73.400	70.000
Q1 2029	197.815	114.884	20.806	94.078	67.591	73.400	72.000
Q2 2029	199.464	112.478	21.231	91.247	65.499	73.400	72.000
Q3 2029	200.714	109.875	21.649	88.226	63.267	73.400	72.000
Q4 2029	201.564	108.184	22.061	86.123	61.708	73.400	72.000
Q1 2030	204.145	107.552	22.543	85.009	73.215	87.552	74.000
Q2 2030	208.647	107.875	23.102	84.773	72.986	87.552	74.000
Q3 2030	213.029	108.051	23.665	84.386	72.625	87.552	74.000
Q4 2030	217.293	108.085	24.232	83.852	72.134	87.552	74.000

Appendix B. full strong redemption cash

Closing bank balances reconcile beginning cash, operating cash flow, cash capex, external financing, bank repayment, redemption, preferred dividends, investment realization and any common issuance.

Quarter	CFO	Capex	External PE	Debt repaid	Redemption	LT sold	End cash
Q4 2026	42.950	25.000	545.000	18.115	0.000	0.000	1,057.528
Q1 2027	40.930	25.000	0.000	18.115	0.000	0.000	1,055.300
Q2 2027	43.801	25.000	0.000	18.115	49.000	0.000	1,006.942
Q3 2027	52.273	25.000	0.000	18.115	0.000	0.000	1,016.056
Q4 2027	63.938	25.000	0.000	3.875	0.000	0.000	1,051.074
Q1 2028	81.759	40.000	0.000	3.875	0.000	0.000	1,088.915
Q2 2028	87.572	40.000	0.000	3.875	0.000	0.000	1,132.568
Q3 2028	88.559	40.000	0.000	3.875	0.000	0.000	1,177.208
Q4 2028	87.664	40.000	0.000	0.000	0.000	0.000	1,224.828
Q1 2029	92.352	35.000	0.000	0.000	0.000	0.000	1,282.136
Q2 2029	94.259	35.000	0.000	0.000	0.000	0.000	1,341.351
Q3 2029	93.980	35.000	0.000	0.000	0.000	0.000	1,400.288
Q4 2029	94.145	35.000	0.000	0.000	686.543	0.000	772.845
Q1 2030	88.755	30.000	0.000	0.000	0.000	0.000	831.556
Q2 2030	90.582	30.000	0.000	0.000	0.000	0.000	892.095
Q3 2030	91.551	30.000	0.000	0.000	0.000	0.000	953.601
Q4 2030	91.897	30.000	0.000	0.000	0.000	0.000	1,015.454

Appendix A. full paced IPO

Probability 9.0%. Funding full; operations paced; exit IPO. All values are USD millions except ownership and shares.

Each leaf calculates operating results, lower-tier claims, cash resources and ownership before any portfolio-weighted per-share calculation.

Quarter	Revenue	GP	Opex	Group OP	Parent OP	Own %	Shares M
Q4 2026	90.000	59.400	14.000	45.400	38.181	85.500	66.500
Q1 2027	98.081	59.132	14.585	44.548	31.705	73.400	68.000
Q2 2027	110.523	66.952	15.328	51.624	36.872	73.400	68.000
Q3 2027	123.488	74.271	16.107	58.163	41.646	73.400	68.000
Q4 2027	138.718	82.931	16.975	65.956	47.339	73.400	68.000
Q1 2028	148.629	86.632	17.665	68.967	49.388	73.400	70.000
Q2 2028	152.256	86.127	18.142	67.986	48.654	73.400	70.000
Q3 2028	154.309	84.193	18.557	65.636	46.917	73.400	70.000
Q4 2028	155.822	81.894	18.961	62.933	44.921	73.400	70.000
Q1 2029	157.152	79.815	19.367	60.448	42.944	73.400	72.000
Q2 2029	158.155	77.530	19.768	57.762	40.960	73.400	72.000
Q3 2029	158.841	76.079	20.167	55.912	39.591	73.400	72.000
Q4 2029	159.211	74.472	20.562	53.910	38.110	73.400	72.000
Q1 2030	161.014	73.663	21.016	52.646	37.027	73.400	74.000
Q2 2030	162.603	72.654	21.472	51.182	35.938	73.400	74.000
Q3 2030	163.990	71.532	21.929	49.603	34.766	73.400	74.000
Q4 2030	165.174	70.307	22.388	47.919	33.517	73.400	74.000

Appendix B. full paced IPO cash

Closing bank balances reconcile beginning cash, operating cash flow, cash capex, external financing, bank repayment, redemption, preferred dividends, investment realization and any common issuance.

Quarter	CFO	Capex	External PE	Debt repaid	Redemption	LT sold	End cash
Q4 2026	42.950	25.000	545.000	18.115	0.000	0.000	1,057.528
Q1 2027	40.117	22.500	0.000	18.115	0.000	0.000	1,056.987
Q2 2027	39.659	22.500	0.000	18.115	49.000	0.000	1,006.986
Q3 2027	43.124	22.500	0.000	18.115	0.000	0.000	1,009.451
Q4 2027	46.455	22.500	0.000	3.875	0.000	0.000	1,029.487
Q1 2028	53.561	30.000	0.000	3.875	0.000	0.000	1,049.129
Q2 2028	58.263	30.000	0.000	3.875	0.000	0.000	1,073.473
Q3 2028	59.476	30.000	0.000	3.875	0.000	0.000	1,099.029
Q4 2028	58.709	30.000	0.000	0.000	0.000	0.000	1,127.694
Q1 2029	63.510	27.500	0.000	0.000	0.000	0.000	1,163.660
Q2 2029	64.510	27.500	0.000	0.000	0.000	0.000	1,200.626
Q3 2029	64.992	27.500	0.000	0.000	0.000	0.000	1,238.073
Q4 2029	65.257	27.500	0.000	0.000	0.000	0.000	1,275.786
Q1 2030	64.129	25.000	0.000	0.000	0.000	0.000	1,314.872
Q2 2030	65.624	25.000	0.000	0.000	0.000	0.000	1,355.452
Q3 2030	65.758	25.000	0.000	0.000	0.000	0.000	1,396.165
Q4 2030	65.450	25.000	0.000	0.000	0.000	0.000	1,436.571

Appendix A. full paced redemption

Probability 6.0%. Funding full; operations paced; exit redemption. All values are USD millions except ownership and shares.

Each leaf calculates operating results, lower-tier claims, cash resources and ownership before any portfolio-weighted per-share calculation.

Quarter	Revenue	GP	Opex	Group OP	Parent OP	Own %	Shares M
Q4 2026	90.000	59.400	14.000	45.400	38.181	85.500	66.500
Q1 2027	98.081	59.132	14.585	44.548	31.705	73.400	68.000
Q2 2027	110.523	66.952	15.328	51.624	36.872	73.400	68.000
Q3 2027	123.488	74.271	16.107	58.163	41.646	73.400	68.000
Q4 2027	138.718	82.931	16.975	65.956	47.339	73.400	68.000
Q1 2028	148.629	86.632	17.665	68.967	49.388	73.400	70.000
Q2 2028	152.256	86.127	18.142	67.986	48.654	73.400	70.000
Q3 2028	154.309	84.193	18.557	65.636	46.917	73.400	70.000
Q4 2028	155.822	81.894	18.961	62.933	44.921	73.400	70.000
Q1 2029	157.152	79.815	19.367	60.448	42.944	73.400	72.000
Q2 2029	158.155	77.530	19.768	57.762	40.960	73.400	72.000
Q3 2029	158.841	76.079	20.167	55.912	39.591	73.400	72.000
Q4 2029	159.211	74.472	20.562	53.910	38.110	73.400	72.000
Q1 2030	161.014	73.663	21.016	52.646	44.937	87.552	74.000
Q2 2030	162.603	72.654	21.472	51.182	43.638	87.552	74.000
Q3 2030	163.990	71.532	21.929	49.603	42.240	87.552	74.000
Q4 2030	165.174	70.307	22.388	47.919	40.750	87.552	74.000

Appendix B. full paced redemption cash

Closing bank balances reconcile beginning cash, operating cash flow, cash capex, external financing, bank repayment, redemption, preferred dividends, investment realization and any common issuance.

Quarter	CFO	Capex	External PE	Debt repaid	Redemption	LT sold	End cash
Q4 2026	42.950	25.000	545.000	18.115	0.000	0.000	1,057.528
Q1 2027	40.117	22.500	0.000	18.115	0.000	0.000	1,056.987
Q2 2027	39.659	22.500	0.000	18.115	49.000	0.000	1,006.986
Q3 2027	43.124	22.500	0.000	18.115	0.000	0.000	1,009.451
Q4 2027	46.455	22.500	0.000	3.875	0.000	0.000	1,029.487
Q1 2028	53.561	30.000	0.000	3.875	0.000	0.000	1,049.129
Q2 2028	58.263	30.000	0.000	3.875	0.000	0.000	1,073.473
Q3 2028	59.476	30.000	0.000	3.875	0.000	0.000	1,099.029
Q4 2028	58.709	30.000	0.000	0.000	0.000	0.000	1,127.694
Q1 2029	63.510	27.500	0.000	0.000	0.000	0.000	1,163.660
Q2 2029	64.510	27.500	0.000	0.000	0.000	0.000	1,200.626
Q3 2029	64.992	27.500	0.000	0.000	0.000	0.000	1,238.073
Q4 2029	65.257	27.500	0.000	0.000	686.543	0.000	589.243
Q1 2030	59.753	25.000	0.000	0.000	0.000	0.000	623.952
Q2 2030	61.219	25.000	0.000	0.000	0.000	0.000	660.127
Q3 2030	61.325	25.000	0.000	0.000	0.000	0.000	696.408
Q4 2030	60.989	25.000	0.000	0.000	0.000	0.000	732.353

Appendix A. minimum strong IPO

Probability 6.0%. Funding minimum; operations strong; exit IPO. All values are USD millions except ownership and shares.

Each leaf calculates operating results, lower-tier claims, cash resources and ownership before any portfolio-weighted per-share calculation.

Quarter	Revenue	GP	Opex	Group OP	Parent OP	Own %	Shares M
Q4 2026	90.000	59.400	14.000	45.400	38.181	85.500	66.500
Q1 2027	103.289	64.642	14.767	49.876	38.790	79.533	68.000
Q2 2027	123.900	79.244	15.797	63.447	49.544	79.533	68.000
Q3 2027	149.552	97.523	17.022	80.501	63.069	79.533	68.000
Q4 2027	175.967	115.836	18.286	97.550	76.594	79.533	68.000
Q1 2028	185.395	121.573	18.963	102.610	80.490	79.533	70.000
Q2 2028	189.687	121.166	19.467	101.699	79.750	79.533	70.000
Q3 2028	193.235	119.475	19.935	99.540	78.017	79.533	70.000
Q4 2028	195.755	116.998	20.375	96.623	75.681	79.533	70.000
Q1 2029	197.815	114.884	20.806	94.078	73.531	79.533	72.000
Q2 2029	199.464	112.478	21.231	91.247	71.264	79.533	72.000
Q3 2029	200.714	109.875	21.649	88.226	68.845	79.533	72.000
Q4 2029	201.564	108.184	22.061	86.123	67.157	79.533	72.000
Q1 2030	204.145	107.552	22.543	85.009	66.142	79.533	74.000
Q2 2030	208.647	107.875	23.102	84.773	65.935	79.533	74.000
Q3 2030	213.029	108.051	23.665	84.386	65.607	79.533	74.000
Q4 2030	217.293	108.085	24.232	83.852	65.161	79.533	74.000

Appendix B. minimum strong IPO cash

Closing bank balances reconcile beginning cash, operating cash flow, cash capex, external financing, bank repayment, redemption, preferred dividends, investment realization and any common issuance.

Quarter	CFO	Capex	External PE	Debt repaid	Redemption	LT sold	End cash
Q4 2026	42.950	25.000	285.000	18.115	0.000	0.000	797.528
Q1 2027	39.273	25.000	0.000	18.115	0.000	0.000	793.642
Q2 2027	42.133	25.000	0.000	18.115	49.000	0.000	743.616
Q3 2027	50.595	25.000	0.000	18.115	0.000	0.000	751.052
Q4 2027	62.248	25.000	0.000	3.875	0.000	0.000	784.381
Q1 2028	80.059	40.000	0.000	3.875	0.000	0.000	820.521
Q2 2028	85.861	40.000	0.000	3.875	0.000	0.000	862.463
Q3 2028	86.837	40.000	0.000	3.875	0.000	0.000	905.381
Q4 2028	85.931	40.000	0.000	0.000	0.000	0.000	951.269
Q1 2029	90.608	35.000	0.000	0.000	0.000	0.000	1,006.833
Q2 2029	92.504	35.000	0.000	0.000	0.000	0.000	1,064.293
Q3 2029	92.214	35.000	0.000	0.000	0.000	0.000	1,121.463
Q4 2029	92.367	35.000	0.000	0.000	0.000	0.000	1,178.786
Q1 2030	91.343	30.000	0.000	0.000	0.000	0.000	1,240.085
Q2 2030	93.187	30.000	0.000	0.000	0.000	0.000	1,303.228
Q3 2030	94.171	30.000	0.000	0.000	0.000	0.000	1,367.355
Q4 2030	94.535	30.000	0.000	0.000	0.000	0.000	1,431.846

Appendix A. minimum strong redemption

Probability 4.0%. Funding minimum; operations strong; exit redemption. All values are USD millions except ownership and shares.

Each leaf calculates operating results, lower-tier claims, cash resources and ownership before any portfolio-weighted per-share calculation.

Quarter	Revenue	GP	Opex	Group OP	Parent OP	Own %	Shares M
Q4 2026	90.000	59.400	14.000	45.400	38.181	85.500	66.500
Q1 2027	103.289	64.642	14.767	49.876	38.790	79.533	68.000
Q2 2027	123.900	79.244	15.797	63.447	49.544	79.533	68.000
Q3 2027	149.552	97.523	17.022	80.501	63.069	79.533	68.000
Q4 2027	175.967	115.836	18.286	97.550	76.594	79.533	68.000
Q1 2028	185.395	121.573	18.963	102.610	80.490	79.533	70.000
Q2 2028	189.687	121.166	19.467	101.699	79.750	79.533	70.000
Q3 2028	193.235	119.475	19.935	99.540	78.017	79.533	70.000
Q4 2028	195.755	116.998	20.375	96.623	75.681	79.533	70.000
Q1 2029	197.815	114.884	20.806	94.078	73.531	79.533	72.000
Q2 2029	199.464	112.478	21.231	91.247	71.264	79.533	72.000
Q3 2029	200.714	109.875	21.649	88.226	68.845	79.533	72.000
Q4 2029	201.564	108.184	22.061	86.123	67.157	79.533	72.000
Q1 2030	204.145	107.552	22.543	85.009	73.215	87.552	74.000
Q2 2030	208.647	107.875	23.102	84.773	72.986	87.552	74.000
Q3 2030	213.029	108.051	23.665	84.386	72.625	87.552	74.000
Q4 2030	217.293	108.085	24.232	83.852	72.134	87.552	74.000

Appendix B. minimum strong redemption cash

Closing bank balances reconcile beginning cash, operating cash flow, cash capex, external financing, bank repayment, redemption, preferred dividends, investment realization and any common issuance.

Quarter	CFO	Capex	External PE	Debt repaid	Redemption	LT sold	End cash
Q4 2026	42.950	25.000	285.000	18.115	0.000	0.000	797.528
Q1 2027	39.273	25.000	0.000	18.115	0.000	0.000	793.642
Q2 2027	42.133	25.000	0.000	18.115	49.000	0.000	743.616
Q3 2027	50.595	25.000	0.000	18.115	0.000	0.000	751.052
Q4 2027	62.248	25.000	0.000	3.875	0.000	0.000	784.381
Q1 2028	80.059	40.000	0.000	3.875	0.000	0.000	820.521
Q2 2028	85.861	40.000	0.000	3.875	0.000	0.000	862.463
Q3 2028	86.837	40.000	0.000	3.875	0.000	0.000	905.381
Q4 2028	85.931	40.000	0.000	0.000	0.000	0.000	951.269
Q1 2029	90.608	35.000	0.000	0.000	0.000	0.000	1,006.833
Q2 2029	92.504	35.000	0.000	0.000	0.000	0.000	1,064.293
Q3 2029	92.214	35.000	0.000	0.000	0.000	0.000	1,121.463
Q4 2029	92.367	35.000	0.000	0.000	359.018	0.000	819.768
Q1 2030	89.054	30.000	0.000	0.000	0.000	0.000	878.778
Q2 2030	90.884	30.000	0.000	0.000	0.000	0.000	939.618
Q3 2030	91.853	30.000	0.000	0.000	0.000	0.000	1,001.427
Q4 2030	92.202	30.000	0.000	0.000	0.000	0.000	1,063.585

Appendix A. minimum paced IPO

Probability 12.0%. Funding minimum; operations paced; exit IPO. All values are USD millions except ownership and shares.

Each leaf calculates operating results, lower-tier claims, cash resources and ownership before any portfolio-weighted per-share calculation.

Quarter	Revenue	GP	Opex	Group OP	Parent OP	Own %	Shares M
Q4 2026	90.000	59.400	14.000	45.400	38.181	85.500	66.500
Q1 2027	98.081	59.132	14.585	44.548	34.562	79.533	68.000
Q2 2027	110.523	66.952	15.328	51.624	40.161	79.533	68.000
Q3 2027	123.488	74.271	16.107	58.163	45.335	79.533	68.000
Q4 2027	138.718	82.931	16.975	65.956	51.503	79.533	68.000
Q1 2028	148.629	86.632	17.665	68.967	53.765	79.533	70.000
Q2 2028	152.256	86.127	18.142	67.986	52.970	79.533	70.000
Q3 2028	154.309	84.193	18.557	65.636	51.088	79.533	70.000
Q4 2028	155.822	81.894	18.961	62.933	48.925	79.533	70.000
Q1 2029	157.152	79.815	19.367	60.448	46.825	79.533	72.000
Q2 2029	158.155	77.530	19.768	57.762	44.675	79.533	72.000
Q3 2029	158.841	76.079	20.167	55.912	43.192	79.533	72.000
Q4 2029	159.211	74.472	20.562	53.910	41.587	79.533	72.000
Q1 2030	161.014	73.663	21.016	52.646	40.455	79.533	74.000
Q2 2030	162.603	72.654	21.472	51.182	39.275	79.533	74.000
Q3 2030	163.990	71.532	21.929	49.603	38.005	79.533	74.000
Q4 2030	165.174	70.307	22.388	47.919	36.651	79.533	74.000

Appendix B. minimum paced IPO cash

Closing bank balances reconcile beginning cash, operating cash flow, cash capex, external financing, bank repayment, redemption, preferred dividends, investment realization and any common issuance.

Quarter	CFO	Capex	External PE	Debt repaid	Redemption	LT sold	End cash
Q4 2026	42.950	25.000	285.000	18.115	0.000	0.000	797.528
Q1 2027	38.460	22.500	0.000	18.115	0.000	0.000	795.329
Q2 2027	37.990	22.500	0.000	18.115	49.000	0.000	743.660
Q3 2027	41.446	22.500	0.000	18.115	0.000	0.000	744.447
Q4 2027	44.766	22.500	0.000	3.875	0.000	0.000	762.794
Q1 2028	51.861	30.000	0.000	3.875	0.000	0.000	780.735
Q2 2028	56.552	30.000	0.000	3.875	0.000	0.000	803.368
Q3 2028	57.754	30.000	0.000	3.875	0.000	0.000	827.203
Q4 2028	56.976	30.000	0.000	0.000	0.000	0.000	854.134
Q1 2029	61.766	27.500	0.000	0.000	0.000	0.000	888.356
Q2 2029	62.755	27.500	0.000	0.000	0.000	0.000	923.567
Q3 2029	63.225	27.500	0.000	0.000	0.000	0.000	959.249
Q4 2029	63.479	27.500	0.000	0.000	0.000	0.000	995.184
Q1 2030	62.340	25.000	0.000	0.000	0.000	0.000	1,032.480
Q2 2030	63.824	25.000	0.000	0.000	0.000	0.000	1,071.260
Q3 2030	63.946	25.000	0.000	0.000	0.000	0.000	1,110.162
Q4 2030	63.626	25.000	0.000	0.000	0.000	0.000	1,148.744

Appendix A. minimum paced redemption

Probability 8.0%. Funding minimum; operations paced; exit redemption. All values are USD millions except ownership and shares.

Each leaf calculates operating results, lower-tier claims, cash resources and ownership before any portfolio-weighted per-share calculation.

Quarter	Revenue	GP	Opex	Group OP	Parent OP	Own %	Shares M
Q4 2026	90.000	59.400	14.000	45.400	38.181	85.500	66.500
Q1 2027	98.081	59.132	14.585	44.548	34.562	79.533	68.000
Q2 2027	110.523	66.952	15.328	51.624	40.161	79.533	68.000
Q3 2027	123.488	74.271	16.107	58.163	45.335	79.533	68.000
Q4 2027	138.718	82.931	16.975	65.956	51.503	79.533	68.000
Q1 2028	148.629	86.632	17.665	68.967	53.765	79.533	70.000
Q2 2028	152.256	86.127	18.142	67.986	52.970	79.533	70.000
Q3 2028	154.309	84.193	18.557	65.636	51.088	79.533	70.000
Q4 2028	155.822	81.894	18.961	62.933	48.925	79.533	70.000
Q1 2029	157.152	79.815	19.367	60.448	46.825	79.533	72.000
Q2 2029	158.155	77.530	19.768	57.762	44.675	79.533	72.000
Q3 2029	158.841	76.079	20.167	55.912	43.192	79.533	72.000
Q4 2029	159.211	74.472	20.562	53.910	41.587	79.533	72.000
Q1 2030	161.014	73.663	21.016	52.646	44.937	87.552	74.000
Q2 2030	162.603	72.654	21.472	51.182	43.638	87.552	74.000
Q3 2030	163.990	71.532	21.929	49.603	42.240	87.552	74.000
Q4 2030	165.174	70.307	22.388	47.919	40.750	87.552	74.000

Appendix B. minimum paced redemption cash

Closing bank balances reconcile beginning cash, operating cash flow, cash capex, external financing, bank repayment, redemption, preferred dividends, investment realization and any common issuance.

Quarter	CFO	Capex	External PE	Debt repaid	Redemption	LT sold	End cash
Q4 2026	42.950	25.000	285.000	18.115	0.000	0.000	797.528
Q1 2027	38.460	22.500	0.000	18.115	0.000	0.000	795.329
Q2 2027	37.990	22.500	0.000	18.115	49.000	0.000	743.660
Q3 2027	41.446	22.500	0.000	18.115	0.000	0.000	744.447
Q4 2027	44.766	22.500	0.000	3.875	0.000	0.000	762.794
Q1 2028	51.861	30.000	0.000	3.875	0.000	0.000	780.735
Q2 2028	56.552	30.000	0.000	3.875	0.000	0.000	803.368
Q3 2028	57.754	30.000	0.000	3.875	0.000	0.000	827.203
Q4 2028	56.976	30.000	0.000	0.000	0.000	0.000	854.134
Q1 2029	61.766	27.500	0.000	0.000	0.000	0.000	888.356
Q2 2029	62.755	27.500	0.000	0.000	0.000	0.000	923.567
Q3 2029	63.225	27.500	0.000	0.000	0.000	0.000	959.249
Q4 2029	63.479	27.500	0.000	0.000	359.018	0.000	636.166
Q1 2030	60.052	25.000	0.000	0.000	0.000	0.000	671.174
Q2 2030	61.520	25.000	0.000	0.000	0.000	0.000	707.650
Q3 2030	61.628	25.000	0.000	0.000	0.000	0.000	744.234
Q4 2030	61.293	25.000	0.000	0.000	0.000	0.000	780.483

Appendix A. delay paced

Probability 5.0%. Funding delay; operations paced; exit no_new_investors. All values are USD millions except ownership and shares.

Each leaf calculates operating results, lower-tier claims, cash resources and ownership before any portfolio-weighted per-share calculation.

Quarter	Revenue	GP	Opex	Group OP	Parent OP	Own %	Shares M
Q4 2026	90.000	59.400	14.000	45.400	38.181	85.500	66.500
Q1 2027	98.081	59.132	14.585	44.548	37.343	85.500	68.000
Q2 2027	110.523	66.952	15.328	51.624	43.362	85.500	68.000
Q3 2027	123.488	74.271	16.107	58.163	48.923	85.500	68.000
Q4 2027	138.718	82.931	16.975	65.956	55.554	85.500	68.000
Q1 2028	148.629	86.632	17.665	68.967	58.024	85.500	70.000
Q2 2028	152.256	86.127	18.142	67.986	57.169	85.500	70.000
Q3 2028	154.309	84.193	18.557	65.636	55.146	85.500	70.000
Q4 2028	155.822	81.894	18.961	62.933	52.821	85.500	70.000
Q1 2029	157.152	79.815	19.367	60.448	50.600	85.500	72.000
Q2 2029	158.155	77.530	19.768	57.762	48.290	85.500	72.000
Q3 2029	158.841	76.079	20.167	55.912	46.695	85.500	72.000
Q4 2029	159.211	74.472	20.562	53.910	44.970	85.500	72.000
Q1 2030	161.014	73.663	21.016	52.646	43.790	85.500	74.000
Q2 2030	162.603	72.654	21.472	51.182	42.522	85.500	74.000
Q3 2030	163.990	71.532	21.929	49.603	41.157	85.500	74.000
Q4 2030	165.174	70.307	22.388	47.919	39.701	85.500	74.000

Appendix B. delay paced cash

Closing bank balances reconcile beginning cash, operating cash flow, cash capex, external financing, bank repayment, redemption, preferred dividends, investment realization and any common issuance.

Quarter	CFO	Capex	External PE	Debt repaid	Redemption	LT sold	End cash
Q4 2026	42.950	25.000	0.000	18.115	0.000	0.000	512.528
Q1 2027	36.643	22.500	0.000	18.115	0.000	0.000	508.512
Q2 2027	36.162	22.500	0.000	18.115	49.000	0.000	455.015
Q3 2027	39.605	22.500	0.000	18.115	0.000	0.000	453.961
Q4 2027	42.914	22.500	0.000	3.875	0.000	0.000	470.456
Q1 2028	49.997	30.000	0.000	3.875	0.000	0.000	486.534
Q2 2028	54.676	30.000	0.000	3.875	0.000	0.000	507.292
Q3 2028	55.866	30.000	0.000	3.875	0.000	0.000	529.239
Q4 2028	55.076	30.000	0.000	0.000	0.000	0.000	554.271
Q1 2029	59.854	27.500	0.000	0.000	0.000	0.000	586.581
Q2 2029	60.831	27.500	0.000	0.000	0.000	0.000	619.868
Q3 2029	61.289	27.500	0.000	0.000	0.000	0.000	653.614
Q4 2029	61.531	27.500	0.000	0.000	0.000	0.000	687.601
Q1 2030	60.380	25.000	0.000	0.000	0.000	0.000	722.936
Q2 2030	61.850	25.000	0.000	0.000	0.000	0.000	759.743
Q3 2030	61.960	25.000	0.000	0.000	0.000	0.000	796.658
Q4 2030	61.628	25.000	0.000	0.000	0.000	0.000	833.242

Appendix A. delay weak

Probability 5.0%. Funding delay; operations weak; exit no_new_investors. All values are USD millions except ownership and shares.

Each leaf calculates operating results, lower-tier claims, cash resources and ownership before any portfolio-weighted per-share calculation.

Quarter	Revenue	GP	Opex	Group OP	Parent OP	Own %	Shares M
Q4 2026	90.000	59.400	14.000	45.400	38.181	85.500	66.500
Q1 2027	90.969	53.126	14.336	38.791	32.438	85.500	68.000
Q2 2027	93.135	51.956	14.720	37.236	31.099	85.500	68.000
Q3 2027	100.708	54.086	15.308	38.779	32.394	85.500	68.000
Q4 2027	111.513	58.479	16.017	42.462	35.513	85.500	68.000
Q1 2028	116.459	58.498	16.529	41.968	34.989	85.500	70.000
Q2 2028	120.003	58.207	17.000	41.207	34.323	85.500	70.000
Q3 2028	123.308	57.658	17.460	40.198	33.446	85.500	70.000
Q4 2028	126.211	56.057	17.913	38.144	31.676	85.500	70.000
Q1 2029	128.722	56.014	18.360	37.653	31.157	85.500	72.000
Q2 2029	130.823	55.678	18.801	36.877	30.480	85.500	72.000
Q3 2029	132.524	55.134	19.235	35.899	29.631	85.500	72.000
Q4 2029	133.826	54.397	19.663	34.733	28.622	85.500	72.000
Q1 2030	136.338	54.244	20.143	34.101	27.980	85.500	74.000
Q2 2030	138.581	54.056	20.622	33.435	27.396	85.500	74.000
Q3 2030	140.565	53.760	21.100	32.660	26.719	85.500	74.000
Q4 2030	142.292	53.362	21.578	31.785	25.956	85.500	74.000

Appendix B. delay weak cash

Closing bank balances reconcile beginning cash, operating cash flow, cash capex, external financing, bank repayment, redemption, preferred dividends, investment realization and any common issuance.

Quarter	CFO	Capex	External PE	Debt repaid	Redemption	LT sold	End cash
Q4 2026	42.950	25.000	0.000	18.115	0.000	0.000	512.528
Q1 2027	37.046	15.000	0.000	18.115	0.000	0.000	516.416
Q2 2027	31.860	15.000	0.000	18.115	49.000	0.000	466.117
Q3 2027	27.459	15.000	0.000	18.115	0.000	0.000	460.417
Q4 2027	26.138	15.000	0.000	3.875	0.000	0.000	467.636
Q1 2028	30.455	18.750	0.000	3.875	0.000	0.000	475.422
Q2 2028	31.443	18.750	0.000	3.875	0.000	0.000	484.196
Q3 2028	31.951	18.750	0.000	3.875	0.000	0.000	493.478
Q4 2028	30.899	18.750	0.000	0.000	0.000	0.000	505.583
Q1 2029	37.089	18.750	0.000	0.000	0.000	0.000	523.878
Q2 2029	39.324	18.750	0.000	0.000	0.000	0.000	544.408
Q3 2029	40.296	18.750	0.000	0.000	0.000	0.000	565.910
Q4 2029	40.745	18.750	0.000	0.000	0.000	0.000	587.861
Q1 2030	39.979	18.750	0.000	0.000	0.000	0.000	609.045
Q2 2030	41.431	18.750	0.000	0.000	0.000	0.000	631.683
Q3 2030	42.000	18.750	0.000	0.000	0.000	0.000	654.889
Q4 2030	42.251	18.750	0.000	0.000	0.000	0.000	678.346

Appendix C. Share and funding controls

The ordinary economic share proxy is 66.5 million in 2026,68 in 2027,70 in 2028,72 in 2029 and 74 in 2030. The initial-share control scales that schedule. Contingent common issuance, if required, affects the following quarter.

The selected group branches require zero emergency common shares. The dilution sensitivity still recalculates the entire share path.

Full precision, branch inputs, asset and cash roll-forwards, source IDs and all formula-control assumptions are retained in AXTI_AXTInc_DeepDive_model.json.